University of Nevada, Reno

Optical Pumping of Rubidium Spin in a Solid Argon Matrix

A thesis submitted in partial fulfillment of the

requirements for the degree of Master of Science in Physics

by

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THE GRADUATE SCHOOL

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prepared under our supervision by

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Abstract

Using the matrix isolation technique we have trapped rubidium atoms inside a cryogenic solid argon crystal of thickness 200 μm. This weakly-interacting solid offers the experimental simplicity and high densities of solid-state systems, while allowing the implanted atoms to retain many of the properties of their gas-phase counterparts. As such, they are a promising environment for quantum information experiments, as well as sensors such as magnetometers.

We spectroscopically observed narrow lines at 799 nm, 785 nm, and 747 nm, and investigated optical pumping of the implanted rubidium atoms. We observed that the absorption spectrum of the rubidium could be modified by incident laser light. This modification could be reversed by applying an additional light source. In addition, we were able to create spin polarization and measure a longitudinal spin relaxation time T_1 of about 0.1 s.

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Chapter 1

Introduction

This thesis deals with the study of spin polarization of rubidium by optical pumping in a solid argon matrix. It has two main parts: chapter 3 describes the preparation of rubidium-doped solid argon matrix; chapter 4 describes the methods used for optical pumping to create spin polarization, and the measurement of longitudinal spin relaxation time. Chapter 2 deals with the apparatus used in the experiment.

Chapter 1 presents the motivation behind the experiment and discusses different atomic spin systems. It also gives an overview of matrix isolation technique. Finally, the optical pumping technique to create spin polarization is discussed in brief.

1.1 Atomic spin systems as qubits

Atomic spin is a desirable candidate for the realization of quantum bit (qubit) because of its simplicity and relative ease of placing an ensemble of atomic spins in a coherent superposition state [9]. For the successful application of a qubit in the field of quantum information there are two key requirements. One is a sufficiently long coherence time, the duration over which a quantum superposition lasts. The other is the ability to initialize and measure the qubit state with high fidelity. In order to improve these

requirements there have been many experimental efforts to create qubits based on spin in different systems of gas and solid phases.

There are three types of existing gas phase spin systems: vapor cells of neutral atoms, trapped neutral atoms, and trapped ions. Atomic vapor cells can achieve long coherence times, on the order of hundreds of seconds [10]. They are also simple experimentally. However, the disadvantage is a low readout fidelity and uncontrolled atomic motion [11, 12, 13]. Trapped neutral atoms and trapped ions are complicated to deal with experimentally. However, with them it is possible to achieve a higher fidelity than with vapor cells, and it is possible to create a number of qubits in a single apparatus. Trapped ions can achieve long coherence times on the order of tens of minutes, and a high readout fidelity of 0.999 has been achieved [14]. Therefore a large number of research groups are working with them. The major disadvantage with trapped ions is that only a limited number of ions can be trapped in a single apparatus.

There are many examples of spin based qubits in solid systems: nitrogen vacancies in diamond, rare-earth ions in crystals [15], phosphorus donor in silicon [16], etc. Solid-state systems offer superior localization of atoms. Their disadvantages, as compared to gas phase systems, are the lower coherence times and the lower initialization and readout fidelities of qubits. Among solid systems, nuclear spin states are attractive options. There have been rapid progress in the ensembles of nuclear spin [17] and a single nuclear spin [18] which offer long coherence times and very high fidelities. However, the problem is the complexity to design and to implement such experiments.

Another attractive alternative is atoms trapped in cryogenic solid matrices of noble gases. Weis and collaborators in 1990's demonstrated the implantation of cesium atoms in solid helium and the optical manipulation of the electron spin state of

implanted atoms [19]. They achieved the longitudinal relaxation times on the order of a second [20] and the transverse relaxation times of at least 0.1 s [21]. Experiments with solid helium are difficult to realize; however, this particular experiment was useful in suggesting the possibility of improving coherence times and fidelity of initialization and readout by using other alkali atoms or host matrices. This will be discussed in the next section.

1.2 Matrix isolated spin systems

The motivation of our experiment on optical pumping of rubidium in solid argon matrix is based on the findings of two key studies. The first is the experiment of 1960's by Kupferman and Pipkin in which they isolated rubidium in solid argon [2]. The second is the experiment of 1990's by Weis and collaborators who demonstrated optical pumping of alkali atoms in solid helium [20, 19]. We will discuss these two experiments in the next subsections.

Matrix isolation is the technique of trapping reactive atoms or molecules in solid crystals of inert gases. This techinique has been used extensively in the chemistry community. The history of cryogenic matrix isolation goes back to 1924 when Kamerlingh Onnes in Leiden studied the emission spectra of oxygen and nitrogen atoms in solid matrices. In 1950's Broida in Washington and Pimentel in Berkeley used this technique to study reactive atoms and molecules [22]. In the physics community, in the 1960's F.M. Pipkin and collaborators at Harvard studied the optical properties of alkali atoms in rare gases including argon [23, 2].

Spin systems based on optical pumping of alkali metals in solid matrices have potential to be applicable in various ways. As pointed out in the section 1.1 such spin systems can be used as qubits in ensemble-based quantum information, especially for quantum networks. It has potential of achieving both of the key requirements for

such applications: long storage lifetimes which requires long coherence time, and high storage and retrieval fidelities which requires high optical densities. Extremely high densities of dopants can be achieved with matrix isolation technique. Additionally, such matrix systems also have the possibility of implementing individual atoms as qubits if the dopant atoms can be isolated at a very low density in the solid matrix.

Such systems also have the potential for constructing a good atomic magnetometer. For a magnetometer the magnetic field noise is given by

$$\delta B = \frac{1}{\gamma\sqrt{nT_2Vt}} \tag{1.1}$$

where γ is the gyromagnetic ratio, n is the density, V is the volume, T_2 is the coherence time, and t is the measurement time [24]. Thus we see that a sensitive atomic magnetometer requires high spin densities and long coherence times, both of which can be expected to be achieved in matrix isolated spin systems. The densities employed by state-of-the-art atomic magnetometers are on the order of 10^{14} cm^{-3}, and at such densities they can attain coherence times on the order of a fraction of a second [24, 25]. With matrix isolated systems we can expect to attain atomic densities on the order of 10^{19} cm^{-3}. Even with short coherence times a sensitive magnetometer could be constructed. Additionally, due to superior localization of atoms they can be used even at very small length scales without affecting coherence time.

1.2.1 Helium matrix isolation

Helium would be an ideal host matrix for a variety of reasons. It does not have a nuclear or an electronic magnetic moment and has the smallest electric polarizability of any atom. The interatomic potential energy of helium is comparable to the zero-point kinetic energy of its condensed phase [19]. This causes its molar volume to become larger than expected, and since the implanted atom can easily repel the surrounding helium atoms, a defect structure called an "atomic bubble" is formed

[19]. This is favorable for spin coherence because the symmetry of the doped atoms will be affected minimally by the helium atoms in the lattice. Unfortunately, such weak interatomic bonds of helium atoms makes the helium lattice so soft that the dopant atoms can diffuse through the crystal forming clusters. The remedy would be to replenish atoms over time, however this is not desirable if we want to use individual atoms as qubits [26].

Another problem is that helium cannot be solidified at absolute zero temperature at low pressure. A pressure of 25 bar is required to solidify helium even at 0 K [19]. As such, we cannot use the standard matrix isolation technique of simultaneously depositing the helium matrix and the dopant atoms on a cold substrate in vacuum. Because of this technical difficulty, helium is not commonly used in cryogenic matrix isolation experiments. To form a doped helium matrix, first a pure helium matrix is created within a pressure cell. Then, by using laser ablation of an atomic target, the matrix is doped with the atoms in the pressure cell [26]. The disadvantage of this method is that there is higher possibility of forming atomic clusters due to ablation process. Additionally, the density of the dopant is rather low: on the order of 10^9 cm^{-3} [26]. With the standard technique, non-helium matrices can be doped at densities on the order of 10^{19} cm^{-3}.

Weis and collaborators successfully performed optical pumping of alkali atoms in solid helium [20, 19]. They found that the pumping mechanism for Cs atoms is primarily by repopulation pumping (in which the spin polarization is not destroyed in the p excited state) [20, 27]. They observed a T_1 time on the order of 1 second in solid helium doped with cesium [20]. They also observed magnetic resonance lines with widths of 10 Hz [21]. This suggests that T_2 times are on the order of 0.1 s, which was confirmed by optical pumping experiments [21].

1.2.2 Argon matrix isolation

In their experiments on isolation of alkali atoms in solid rare gas matrices, Pipkin and collaborators obtained high quality spectra, although the matrices they formed were not of high transparency. The optical depths from the atoms they trapped were on the order of 1. Their objective was to orient the atoms by optical pumping or by using a high magnetic field [23]. However, they did not observe successful optical pumping.

They performed experiments on isolation of rubidium atoms in solid argon crystal. They formed the crystals by depositing a beam of mixture of Rb and Ar onto a cold substrate. Typical concentration of Rb in Ar was in the range 1 in 1000 to 1 in 2000. They detected Rb atoms by absorption spectroscopy which showed two groups of lines broadened by tens of nanometers: the red group having triplet structure, and the blue group having four lines. The red group lines (798.1±2.5 nm, 774.8±1.4 nm, 753.9±1.5 nm) were near to those of the free Rb atom while the blue group lines (713.0±1.1 nm, 696.1±1.1 nm, 678.8±1.3 nm, 656.6±1.0 nm) were shifted about 1650 cm^{-1}. The presence of two groups of lines suggests that there are two types of trapping sites [2].

Although they did not detect optical pumping they induced a polarization by applying a magnetic field on the order of thousands of Gauss. They then observed the differential absorption of σ^+-polarized and σ^--polarized light. This suggested that the optical manipulation and detection of spin should be possible. They found that at 797.9±2.5 nm the differential absorption of polarized light was highest. Figure 3.11 shows the spectrum obtained in their experiment which can be compared to the spectrum from our experiment as shown in figure 3.12.

Noble gases are typically used as host matrix because they interact weakly with the dopant. Helium is avoided due to the difficulties described in section 1.2.1. Neon has the advantage of having weakest bonds among the non-helium noble gases; however

it suffers from having naturally occuring isotopes with nuclear spin. Argon has the second weakest bonds among the non-helium noble gases, and it has no naturally occurring isotopes with nuclear spin [28]. In other systems the fluctuating nuclear spin has found to limit the coherence time [29].

Regarding the dopant, alkali atoms are chosen for optical pumping due to the presence of a single valence electron. We avoid atoms having orbital angular momentum since the anisotropic electronic wavefunction of such atoms could strongly couple to the matrix, and this can lead to short coherence times [30]. Also we avoid $L = S = 0$ atoms since broadening is expected to be too large to directly optically address the nuclear spin states. Among the alkali atoms lithium, rubidium, and cesium are good candidates. In our experiment we choose rubidium and argon because of the availability of past studies of rubidium in solid argon, as described above.

1.3 Optical pumping

Optical pumping is a technique in which the interaction of polarized light causes an alteration of population of energy levels that is different from the thermal equilibrium [15]. Let us consider a simplified atomic system which has a ground state $|g\rangle$, an excited state $|e\rangle$, and a metastable state $|m\rangle$, as shown in figure 1.1. If we apply a polarized electromagnetic radiation to an ensemble of atoms it will excite them from the state $|g\rangle$ to $|e\rangle$. Suppose it will not excite atoms from $|m\rangle$ to $|e\rangle$ due to selection rules, and therefore this metastable state can be called a "dark state". Then during spontaneous decay the atoms will decay from $|e\rangle$ to both states $|g\rangle$ and $|m\rangle$. Thus after a continuous excitation and decay process the atoms be will be eventually populated in the state $|m\rangle$, resulting in an optical pumping [31].

In our experiment, we optically pump rubidium which can be modeled like a hydrogen-like system since it has a single electron in its valence shell. As described in

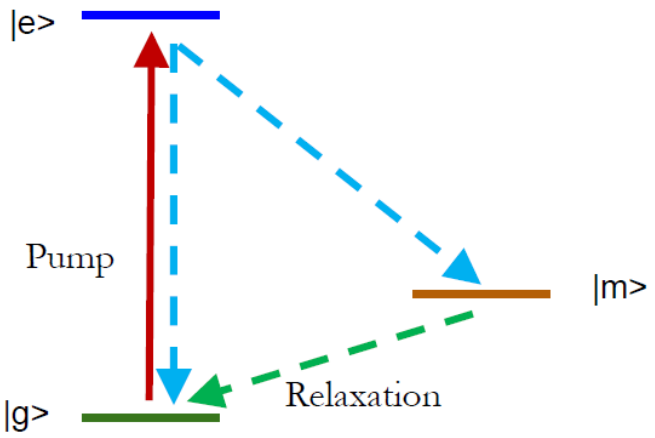


Figure 1.1: Optical pumping process.

chapter 4, we pump rubidium atoms using $5s\ ^2S_{1/2} \rightarrow 5p\ ^2P$ transition. Rubidium has two naturally occurring isotopes: ^{85}Rb (72.17%) and ^{87}Rb (27.84%). The hyperfine structure energy level diagrams for both are presented in appendix C.

When a weak uniform magnetic field is applied, the splitting of hyperfine levels occur due to Zeeman effect. In a weak magnetic field, the Zeeman splitting is given by

$$\Delta E = g_F \mu_B m_F B_z \tag{1.2}$$

where g_F is the Landé g-factor, $\mu_B = \frac{e\hbar}{2m_e}$ is the Bohr magneton, m_F is the spin projection, and B_z is the z-projection of the magnetic field [31]. Thus in weak magnetic field, the splitting is directly proportional to B.

The transition strengths for ^{85}Rb(I = 5/2) due to σ^+ light has been presented in figure 1.2. If a σ^+ pumping light is applied then only the transitions to an excited state with Δm_F =+1 are allowed by selection rule. Therefore, during optical pumping, one of the Zeeman sublevels can act as the dark state [1]. After a continuous excitation and decay the overall population of the atoms will be driven towards the dark state,

[1]We note that in our experiment, due to mixing of the $^2P_{1/2}$ and $^2P_{3/2}$ excited states we do not have a perfect dark state.

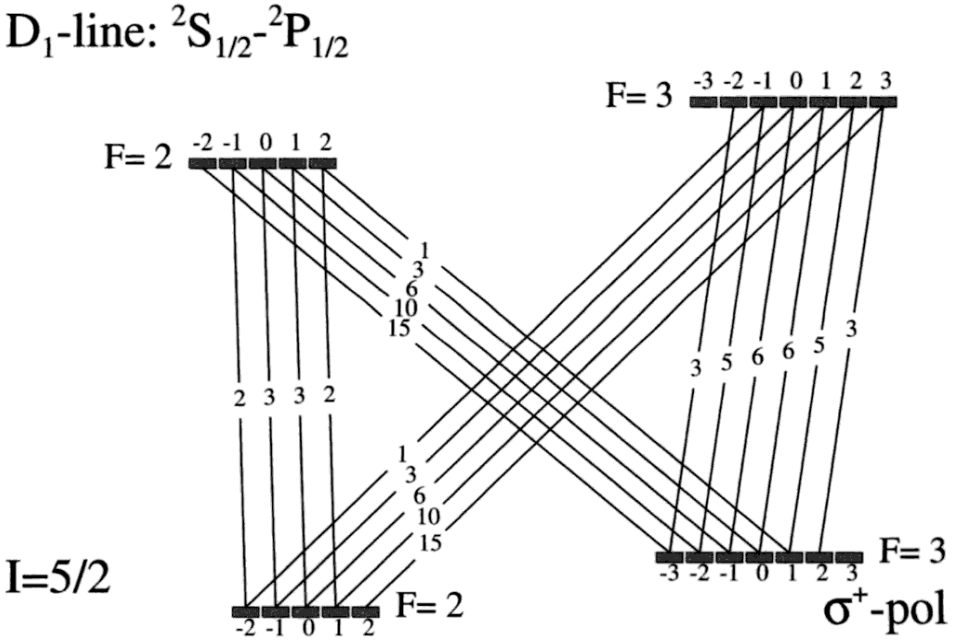


Figure 1.2: Transition strength for ^{85}Rb for optical transitions using σ^+-polarized light from the ground state $5s\ ^2S_{1/2}$ to the first excited state $5p\ ^2P_{1/2}$. We can see that for the state $F = 3$, $m_F = 3$, there is no transition to the excited state and it acts as a dark state. Figure reproduced from p. 286 of reference [32].

resulting in an optical pumping and net spin polarization in the ensemble of Rb atoms. This will cause a decrease in absorption of σ^+ light and increase in absorption of σ^- light. If the pumping light is stopped, then the population of the atoms will be driven back to thermal equilibrium and the spin polarization vector will relax over time. This can be detected by a return in the absorption of σ^+ and σ^- to their equilibrium values.

Two time constants are important: the longitudinal relaxation time (T_1) and the transverse relaxation time (T_2). These times can be measured by using optical pumping and polarization absorption spectroscopy. If the magnetic field is applied longitudinally the atoms precess around this direction. Then if a σ^+ polarized pumping beam is applied along the same direction, the atoms are pumped to a dark state, as described above, forming a net spin polarization. Then when the pumping beam is turned off, the net spin polarization vector relaxes, and the time constant for such

relaxation is called the T_1 time. The experiment described in this thesis is limited to measurement of T_1 time. A typical measurement of T_1 is presented in figure 4.14.

If the magnetic field is applied in transverse direction of the σ^+ pumping light then atoms are placed into a superposition of m_F states, and the spin polarization vector undergoes oscillation as it precesses around the direction of magnetic field. When the pump beam is turned off this oscillation relaxes, and the time constant for such relaxation is called the T_2 time.

Chapter 2

Apparatus

For the successful realization of optical pumping of rubidium in solid argon matrix we need to design a system that is capable of creating argon crystals doped with rubidium at cryogenic temperatures. In this chapter the design of the parts used for this purpose will be described. Furthermore, other tools like sensors, helmholtz coils, data acquisition systems, etc., will be described.

2.1 Cryogenic parts

2.1.1 Cryostat

We have an existing cryostat built for the purpose of buffer-gas cooling experiments. It consists of a pulse tube cooler PT405 available from Cryomech [33] which is currently capable of cooling to about 3.2 K. An aluminum cylinder of diameter 12" serves as a vacuum chamber to maintain pressure equal to or less than 1×10^{-5} torr. Such vacuum of very low pressure is essential in order to restrict the convection of heat as well as to prevent our rubidium sample and argon-rubidium (Ar-Rb) matrix from being impure. The pressures inside the chamber, the oven (described in subsection

2.1.2), and a vacuum pump-manifold are measured by three vacuum transducer gauges from MKS Instruments [34]. The sensors can measure pressures as low as 1×10^{-5} torr but not below that limit. A residual gas analyzer RGA100 from Stanford Research Systems [35] is used to leak-check the cryostat with helium.

Inside the aluminum vacuum chamber there is an aluminum shielding box which encloses the parts needed for matrix isolation. This box wrapped with layers of aluminized mylar sheets helps to block the blackbody radiation due to room temperature. The outer chamber as well as the inner shielding box both have five windows each of diameter 3" and thickness 0.5" in order to have optical access. Cryogenic silicon diode sensors DT-670 [36], which are calibrated relative to [37], are attached to different parts of the cryostat in order to monitor temperature. Details about the design of the cryostat with cooling and vacuum system as well as windows and other accessories have been well explained in a previous dissertation [1].

To serve the purpose of our experiment of matrix isolation we modified the cryostat by removing an existing cell (which was used for buffer-gas cooling experiments in the past) and adding parts as described in the next subsections. Figure 2.1 shows the internal view of the cryostat along east-west direction, with the vacuum chamber and aluminum shielding box removed.

2.1.2 Oven, cover, and fill line

An oven and its cover were designed in order to contain rubidium. It is made of stainless steel and is cube-shaped, with length, width, and height of 1" each. Internally the oven has a cylindrical cavity of diameter 0.75" and depth also of 0.75". A cover is sealed with the oven using a teflon O-ring, and held tight with 4-40 UNC screws. The middle of the cover has a hole of diameter 0.124" which is used to connect the oven with the fill line. Figure 2.2 shows the oven with nozzle, fill line, a resistor heater,

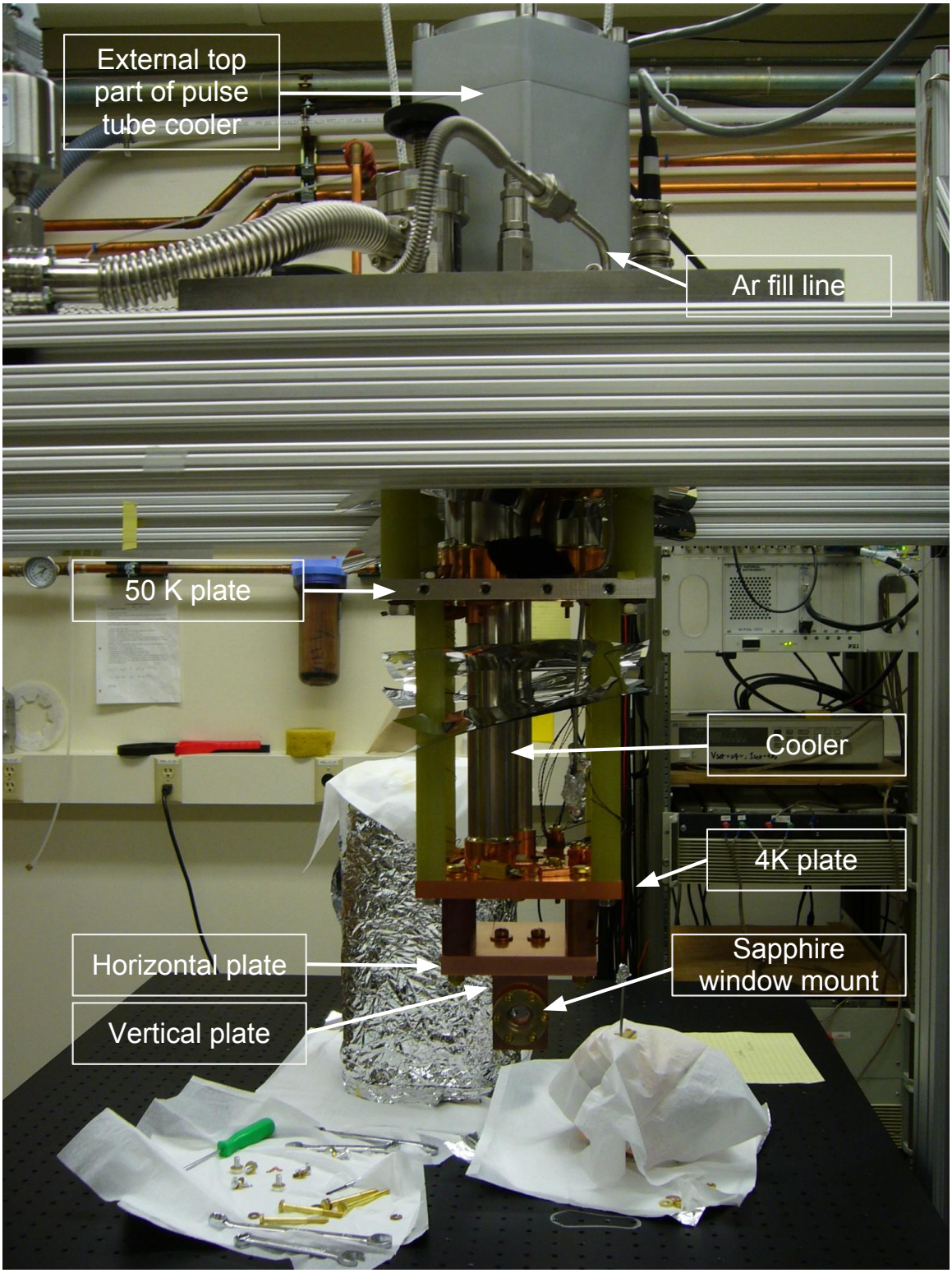


Figure 2.1: Internal view of the cryostat (oven not shown) along east-west direction, with the vacuum chamber and aluminum shielding box removed.

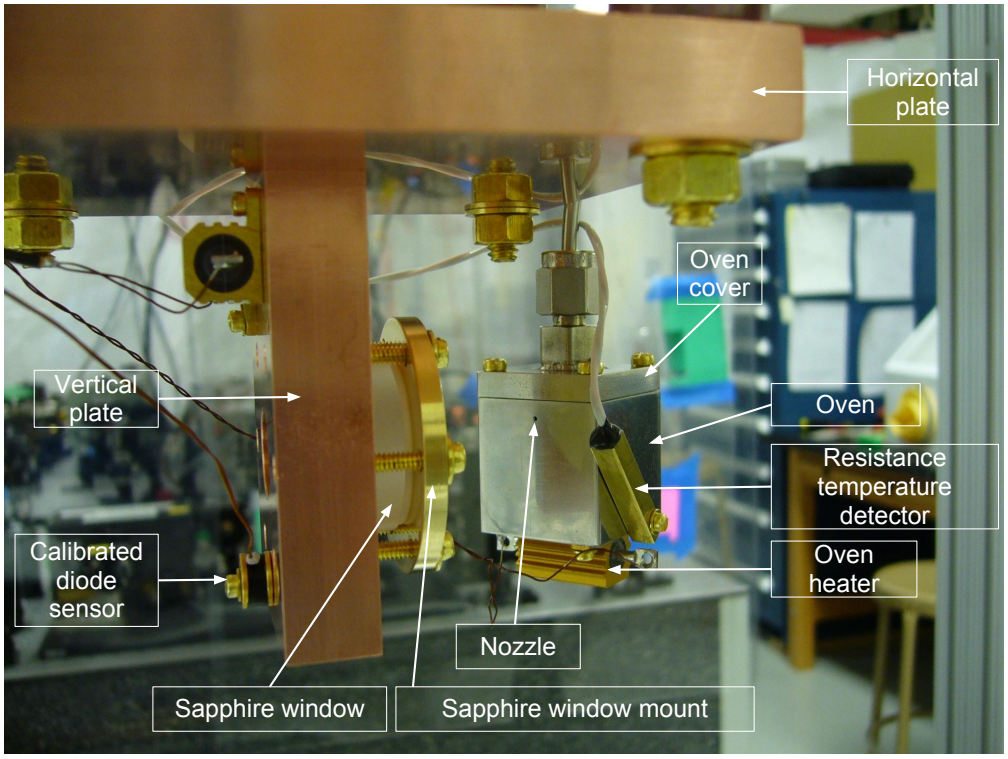


Figure 2.2: Oven with nozzle on the side facing the sapphire window.

and a temperature detector. Appendix A presents the mechanical drawings of the oven and cover.

A resistor heater of 100 Ω [38], which can be attached to the base or a side of the oven, is used to heat the rubidium sample contained inside the oven so that the amount of rubidium deposited can be controlled by adjusting the temperature. In order to monitor the oven temperature a flat platinum resistance detector [39] is attached to one side of the oven.

The side of the oven facing the sapphire window has a clear hole of diameter 0.020". This hole acts as a nozzle for the flow of argon with rubidium during deposition. As can be seen in figure 2.3, the oven is placed slightly right of the perpendicular axis of sapphire window for the purpose of clear optical access to the window. The nozzle side of oven is making an angle $15 - 20^\circ$ with the plane of the sapphire window. The

distance between the nozzle and the sapphire window is about 2".

To control the right amount of flow of rubidium and argon gas during formation of an Ar-Rb crystal, we need to design the correct diameter of the nozzle of the oven. Below we present the calculation for the nozzle diameter.

We assume that 100 μm thick Ar-Rb crystal can be formed in 1 hour and that the ratio of number of rubidium atoms to argon atoms will be 1 to 1000. Then the ratio of argon pressure (P_{Rb}) to rubidium pressure (P_{Ar}) should be $\frac{P_{Rb}}{P_{Ar}} = \frac{1}{1000}$. If we assume that $P_{Ar} = 10$ torr, then we have $P_{Rb} = 1 \times 10^{-2}$ torr. From the rubidium vapor pressure curve in appendix B we see that the required oven temperature to get 1×10^{-2} torr of rubidium pressure is $T_{oven} = 440$ K.

Next, we find the throughput of argon to deposit 100 μm thick Ar-Rb crystal on the sapphire window. We assume that during deposition the Ar-Rb beam will be sprayed into 2π steradians. Then at a distance of 2", the total volume of the argon solidified would be 1.6×10^{12} μm^3. The density of solid argon is 1.768 g cm^{-3} at 20 K [40], and therefore the mass of solidified argon is 2.9 g. With argon atomic weight 40 and Avogadro's number 6.022×10^{23} we find that the number of argon atoms solidified is $N_{Ar} = 4.3 \times 10^{22}$. Then using the ideal gas law we find $P_{Ar} \cdot V_{oven} = N_{Ar} \cdot K \cdot T_{oven} = 2.0 \times 10^3$ torr liters at 440 K. Then assuming that argon deposition rate is 100 μm per hour, we find the throughput of argon to be $Q_{Ar} = 0.56$ torr liter/sec.

Next, we find the conductance of the oven nozzle, C_n. The throughput is given by [41]

$$Q_{Ar} = (P_{Ar} - P_n)C_n \qquad (P_{Ar} > P_n), \tag{2.1}$$

where P_n is the pressure at the end of nozzle which we assume to be zero. From this we see that $C_n = \frac{Q_{Ar}}{P_{Ar}} = 5.6 \times 10^{-2}$ liter/sec.

The mean free path of argon, at the pressure of 10 torr and at room temperature,

is $\sim$5 μm [41], which is much smaller than the nozzle diameter, and therefore the flow of gas in our system is viscous rather than molecular. We want the viscous flow to be in laminar and not turbulent regime. In laminar regime, at 20 °C, the conductance of a long tube having a circular cross section is given by [41, 42]

$$C_n = 180\frac{{D_n}^4}{L_n}P_{Ar} \text{ liter sec}^{-1} \tag{2.2}$$

where D_n is the diameter (in cm) and L_n is the length of the nozzle (in cm), P_{Ar} is the average argon pressure in oven (in torr).

Then with $L_n = 0.318$ cm, $C_n = 5.6 \times 10^{-2}$ liter/sec, and $P_{Ar} = 10$ torr, equation 2.2 gives us $D_n = 0.06$ cm (0.02") for the nozzle. Therefore, we have designed the oven with nozzle having diameter of 0.020" with tolerance $\pm$0.005 as shown in figure A.1 of appendix A.

A pressure gauge is attached to the fill line and we use the pressure there (P_g) to get information about oven pressure(P_{Ar}), assuming $P_g \approx P_{Ar}$. This assumption will be true when $C_n \ll C_f$, where (C_f) is the conductance of the fill line. The fill line has an inner diameter of $\sim$0.3 cm and length of about 50 cm. The nozzle has the diameter of 0.06 cm and length of 0.125 cm. Then using these values in equation 2.2 we see that $\frac{C_f}{C_n} = 2\frac{P_g+P_{Ar}}{P_{Ar}} \geq 4$. This suggests that our measurement of oven pressure could be off by at most 25%.

2.1.3 Horizontal plate, vertical plates, and sapphire window holder

In order to hold the sapphire window we designed a horizontal plate, a vertical plate, and a holder which are shown in figure 2.3. Appendix A shows their mechanical drawings.

The horizontal plate is a square shaped slab made of copper (110) having length

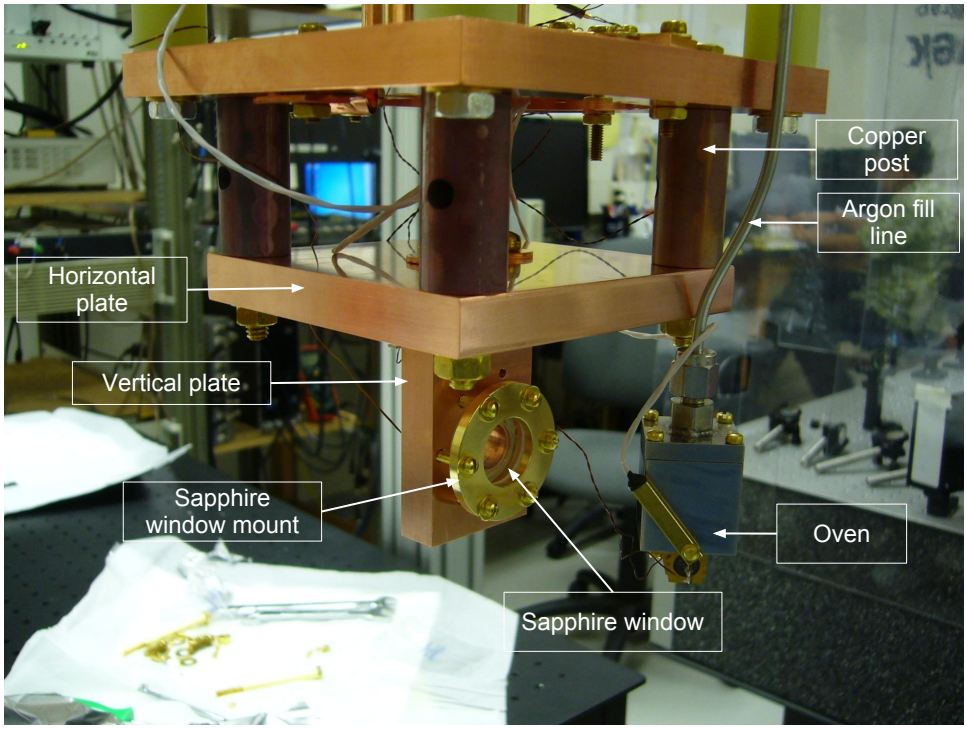


Figure 2.3: Vertical and horizontal plates, sapphire window holder, oven, and fill line, all assembled together.

and width of 4" and thickness of 0.50". Using four brass bolts this horizontal plate is attached to four copper posts which are also attached to another copper plate above. The latter copper plate is attached thermally to the second stage of the pulse tube cooler which has been explained in a previous dissertation [1]. The horizontal plate is anchored to the vertical plate with a pair of brass bolts and copper washers.

The vertical plate is made of copper (110) and has height 2.5", width 1.5" and thickness 0.5". At the center of this vertical plate there is a through hole of diameter 0.5" on top of which sapphire window is mounted. Around this central hole there are six blind tapped holes of depth 0.250" which are used to anchor the plate to the sapphire window holder using 4-40 UNC screws. During the experiment it is often necessary to change the temperature of the sapphire window. Therefore, a resistor heater [38] to change the temperature and a calibrated cryogenic silicon diode sensor

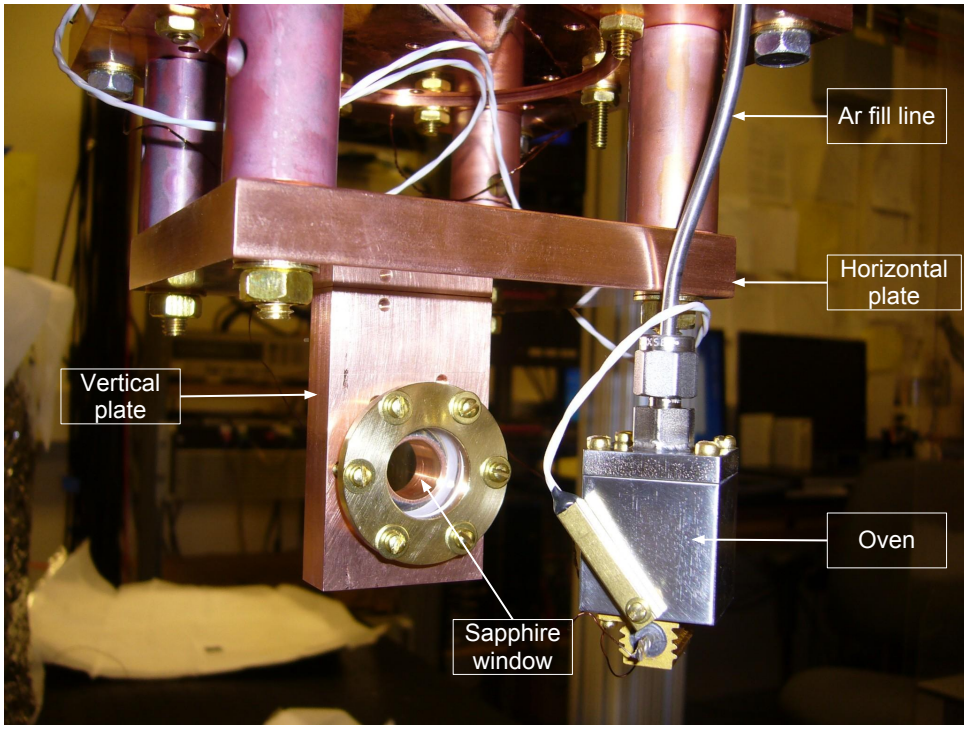


Figure 2.4: Vertical and horizontal plates, sapphire window, oven, and fill line, all assembled together.

DT-670-CU-1.4D [37] to measure the temperature are attached on one side of the vertical plate.

In order to attach the sapphire window to the vertical plate, a flat ring-shaped holder was designed. It has an outer diameter of 1.5" and an inner diameter of 0.750". Six through holes of diameter 0.120" are drilled in order to use 4-40 UNC screws to attach the holder to the vertical plate.

2.1.4 Sapphire window

For the purpose of depositing argon with rubidium we need a substrate. In our experiment we use a sapphire precision window WG31050 from Thor Labs [43] having diameter 1" and thickness 5.0 mm. We chose sapphire as a substrate because of its high thermal conductivity [44]. It has high transmission in the range of 150 nm to 5

μm, and is chemically inert [43]. Its exceptionally hard surface is useful since we need to clean the sapphire window when it becomes dirty after deposition of argon. It is placed between the holder and the vertical plate and is held tight by using screws as described in section 2.1.3. An indium ring is placed between the sapphire window and the vertical plate for thermal contact. A teflon O-ring is placed between the sapphire window and the holder to avoid direct contact with the metal which might cause cracking. The sapphire window is birefringent as discussed in section 4.2.

2.1.5 Photodetectors, spectrometer and cameras

To detect the differential absorption of σ^+ and σ^- components of probe light we use two home-built photodectors which consist of silicon photodiodes S1336-18BQ from Hamamatsu [45], and op-amp OPA2743 from Texas Instruments [46] or AD822AN from Analog Devices [47].

In order to take spectra of the Ar-Rb crystal during the growth of crystal and afterwards we use a miniature grating spectrometer from Ocean Optics [48] that has a wavelength range of 350-1000 nm. It has a entrance slit of 25 μm and has optical resolution of 2.0 nm [48]. We use a halogen lamp or an incandescent lamp as a light source for the spectrometer.

To detect the position of pump and probe beams on the Ar-Rb crystal and to ensure their overlapping we use a CCD camera Flea2 (TM) (FL2-03S2M) [49] from Point Grey. Also, we use a CCD camera connected to a monitor to regularly monitor the status of the Ar-Rb crystal on the sapphire window.

2.2 Helmholtz coils

In order to optically pump and observe polarization of rubidium ensemble in argon crystal we need to apply a weak uniform magnetic field along the direction of a pumping light, as discussed in the section 1.3. We achieve this by using three pairs of Helmholtz coils in east-west, north-south and up-down directions. The coils are attached to the outer chamber of the cryostat. They are made of copper wires and their details are presented in table 2.1.

Coils	Radius [inch]	Turns/coil	Calculated [G/A]	Prior measured [G/A]	Measured [G/A]
North-South	11.3	10	0.32	0.33	0.20
East-West	12.3	10	0.29	–	0.20
Up-Down	6.8	5	0.26	0.28	–

Table 2.1: Helmholtz coils used in the experiment and their strengths. All values are from a previous dissertation [1] except the "measured" values.

2.3 Chemicals

Since one of the main parts of our experiment is to optically pump the rubidium atoms in solid argon, purity of samples is very important. We use 99.998% pure compressed argon gas contained in a cylinder supplied by Praxair [50]. In order to purify argon further by isolating traces of impurities like water, we use dry ice and ethanol to cool a part of the argon pipeline while growing argon crystal as described in chapter 3.

As for the rubidium sample to be trapped in argon crystal we use 99.75% pure rubidium metal obtained from Alfa Aesar ampouled under argon [51]. Rubidium is

highly reactive in air, and therefore the ampoule has to be broken inside an air-tight glove box filled with argon in order to transfer it to the oven. The purity of rubidium may become lesser while breaking the ampoule and transferring rubidium to the oven due to some possibility of oxidization.

2.4 Data acquisition and analysis

We control the opening and closing of shutters and thus the pump beam duration by using transistor-transistor logic (TTL) signal sent to a shutter driver. We use a data acquisition system built with the NI PXIe-6366 [52] card which consists of digital output (DO) and analog input (AI). In a Labview program [52] an user can input arbitrary time steps and sequences, and the timing signal is sent to a the DO part of the NI PXIe-6366 board by direct memory access (DMA). The DO has an internal clock scan rate of 10 MHz which is fast enough for our purpose. It synchronizes the AI part of the NI PXIe-6366 board so as to acquire signals from the photodetectors. Figure 4.12 of chapter 4 presents an example of timing control in our experiment. The details of data acquisition system in our lab has been explained in a previous dissertation [1]. We use the system to monitor the crystal growth as well as for optical pumping experiments.

We use a Labview program to view and acquire data about cryostat temperature and photodiode signals. For data analysis we use IGOR Pro software which can be programmed to load and display data automatically [53].

Chapter 3

Matrix Isolation

In this chapter we present the process used to grow "blank" argon crystals (without rubidium) and rubidium-doped argon crystals. We also discuss monitoring the growth of crystals during deposition, and describe the different kinds of crystals formed under various conditions. The spectrum of rubidium trapped in an argon crystal will be described and compared with that of a past experiment. Finally, analysis of the density of rubidium in the crystal will be presented.

3.1 Growing blank argon crystals

Before we grew rubidium-doped argon crystals we grew many blank argon crystals at different oven and sapphire window temperatures, and at different oven pressures. Growing blank crystals at different conditions would give us an idea about how thick and transparent a crystal we can achieve, and how we can improve the quality of the crystals formed under different conditions. In the course of the experiment we have grown a total of 50 argon crystals (with and without rubidium).

3.1.1 General procedure of argon deposition

First, the cryostat chamber is pumped out to a pressure $< 1 \times 10^{-3}$ torr, and then a residual gas analyzer [35] is used to leak check the cryostat with helium. Then we start the pulse tube cooler [33] which begins to cool down the cryostat. With this cooler a minimum temperature of about 3.2 K can be achieved at the vertical plate. By controlling a resistor heater [38] attached to the vertical plate we can raise the temperature at which we desire to grow an argon crystal. With oven temperature below 20 °C we expect essentially no rubidium in the argon gas beam. If we want to grow a crystal doped with rubidium we heat the oven with another resistor heater [38] to a proper temperature suggested by the rubidium vapor-pressure curve as shown in appendix B. Even to grow a blank crystal we may need to heat the oven so as to unclog the nozzle to facilitate argon flow. We monitor the temperature of the vertical plate to which the sapphire window is fixed with the help of a calibrated diode [37], the temperature of oven using a thermistor [39], and the pressures of the oven, chamber and the manifold using Pirani gauges [34], as described in chapter 2.

Once the temperature of the sapphire window and the oven is stabilized to the desired level, we let argon flow into the oven by crack-opening the oven valve to the argon fill line. The oven pressure is typically maintained between 10 to 40 torr, and argon (along with rubidium if oven is heated to a required temperature) begins to deposit on the sapphire substrate. The crystal growth is monitored with thin film interference as described in subsection 3.1.2. After growing a crystal to the desired thickness, the oven valve is closed. If we decide to keep the crystal, the oven heater is turned off and the sapphire window is cooled very slowly down to 3.2 K. If we decide to remove the argon crystal and clean the sapphire window, we can follow the process as described in subsection 3.1.3 below.

3.1.2 Monitoring the crystal growth

We use thin film interferometry to monitor the growth and measure the thickness of the crystal. Additionally, we can inspect the status of a crystal (whether it is clear or cloudy) by eye and also by using a CCD camera connected to a monitor. The optical setup for monitoring the deposition of argon is shown in figure 3.1. As a light source we use a red LED (LED631E) [54] which has 635 ± 10 nm as the center wavelength of light emitted. A 75 mm plano-convex lens roughly collimates the LED beam which is passed through the growing crystal and is then guided to a photodetector. Since the oven nozzle is aiming at the sapphire window at an angle $\sim 15 - 20\,^{\circ}$, the crystal formed would be of uneven thickness. Therefore, the iris diameter is set to the minimum of about 500 microns in order to monitor the smallest possible portion of crystal growing. We place iris such that it is in the image plane of the sapphire window formed by a biconvex lens of 150 mm. A bandpass filter with central wavelength of 632.8 ± 0.2 nm and full width at half maximum (FWHM) of 1 ± 0.2 nm [55] is placed in front of the photodetector in order to block the room lights and increase the coherence length.

The rate of argon deposition can be controlled by controlling the pressure of argon in the oven which in turn is adjusted by a valve in the fill line. As described in chapter 2, the oven is placed slightly towards the right of the perpendicular axis of the sapphire window, but the nozzle is aimed at the sapphire window. When argon deposits on the sapphire window layer by layer, then due to thin-film interference the photodetector signal shows oscillation over time, as shown in figure 3.5 and 3.6. For a beam normal to the crystal, the interference will be destructive under the condition

$$2n_{Ar}t = m\lambda, \tag{3.1}$$

where n_{Ar} is the refractive index of solid argon (which is 1.26820 ± 0.00016 at 83.81 K [56]), t is the thickness of the crystal layer, m is the number of fringes, and λ is the

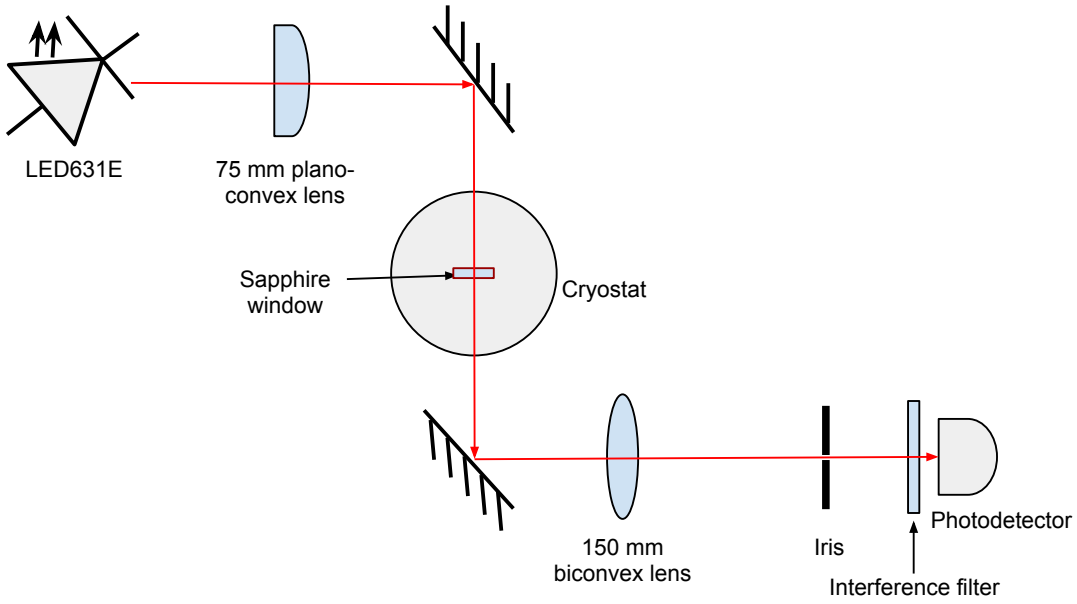


Figure 3.1: Schematic of the optics setup to monitor the growth of a blank argon crystal.

wavelength of LED light. Then the thickness of the argon crystal is given by $t = \frac{m\lambda}{2n_{Ar}}$.

Figures 3.5 and 3.6 present a typical interference fringes observed using an LED. Furthermore, we also used laser at wavelength 799 nm and observed the interference fringes as presented in figure 3.7. In this experiment, we have grown crystals with thickness ranging from about 30 to 200 microns. Due to uneven thickness of the crystal the amplitude of fringes disappear after some time, as can be seen in figure 3.5. However the frequency of fringes gives us the rate of deposition which can be extrapolated to measure the total thickness.

3.1.3 Cleaning the sapphire window for next deposit

If we decide to remove the crystal we turn off the pulse tube cooler so that whole cryostat along with the sapphire window starts to warm up. Also, we keep heating the sapphire window until the crystal melts away, usually around 50 K. Melting can be inspected by eye, by using a CCD camera, and also by monitoring the photodiode

signal.

After removing the argon crystal we may still see some dirt on the window which usually cannot be removed by heating. As described in subsection 3.1.5 below, trying to deposit argon on top of this dirty sapphire window does not result in a good transparent crystal. But if we warm up the cryostat to room temperature, we can grow another good crystal on this slightly dirty sapphire window. This warming up of cryostat to room temperature takes about 2 days if the chamber is left being pumped. But if it is left unpumped and evaporated argon is let to remain in the cryostat, then due to better convection it takes only about 1 day to warm up to room temperature.

Due to repeated use of the sapphire window it may become too dirty in that we can not grow any more good crystals. Figure 3.2 shows the image of one of the cloudy crystals with a few patches of clear parts visible. Furthermore, sometimes cloudy dirt may stick on to the sapphire window and may not fall while warming up. When this happens, we need to open up the cryostat and clean the sapphire window by hand using lens-cleaning paper and ethanol or acetone. If the oven nozzle is clogged, or if we suspect that rubidium in the oven is oxidized we may also open the oven and clean its inside as well as the nozzle and cover by hand. Then we can load rubidium from a new ampoule and set up the oven back to the cryostat for next growth of crystal. Figure 3.3 shows oxidized rubidium clogging the oven nozzle.

3.1.4 Growing argon crystal below 5 K

We tried to grow our first argon crystal with the sapphire temperature at 3.2 K without heating the sapphire window and oven. But this trial to deposit argon failed due to a clog in the oven nozzle. Then we heated the oven to 278 K, and we were able to have a flow of argon, but the deposit on the sapphire window was cloudy. The clog could be due to some oxides of rubidium which may have formed when we

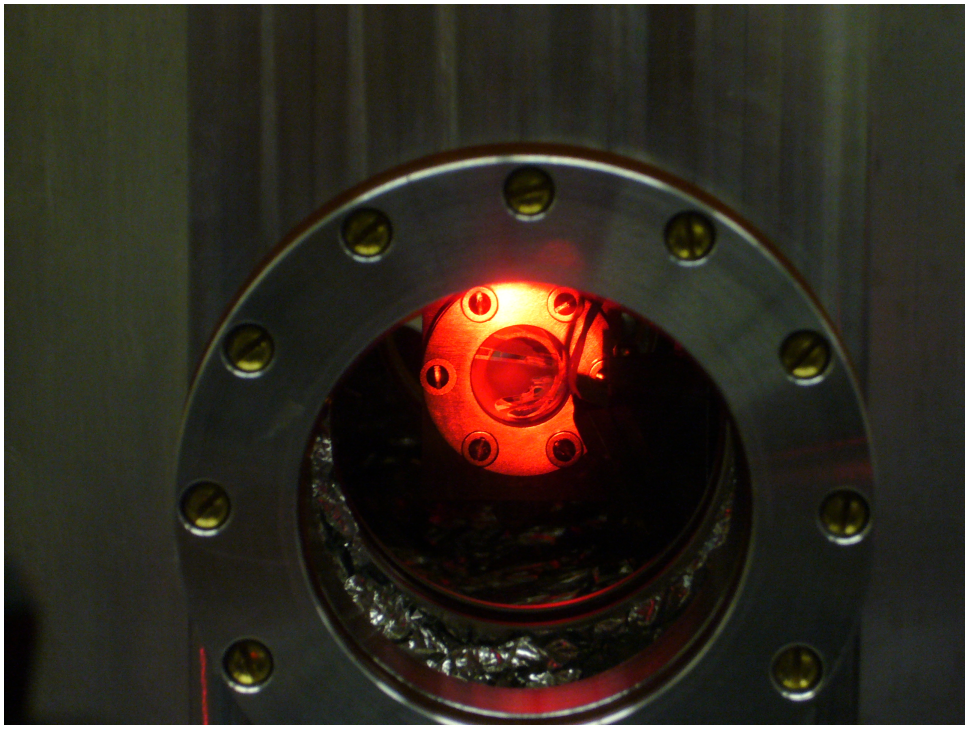


Figure 3.2: Image of a cloudy crystal with a few patches of clear parts.

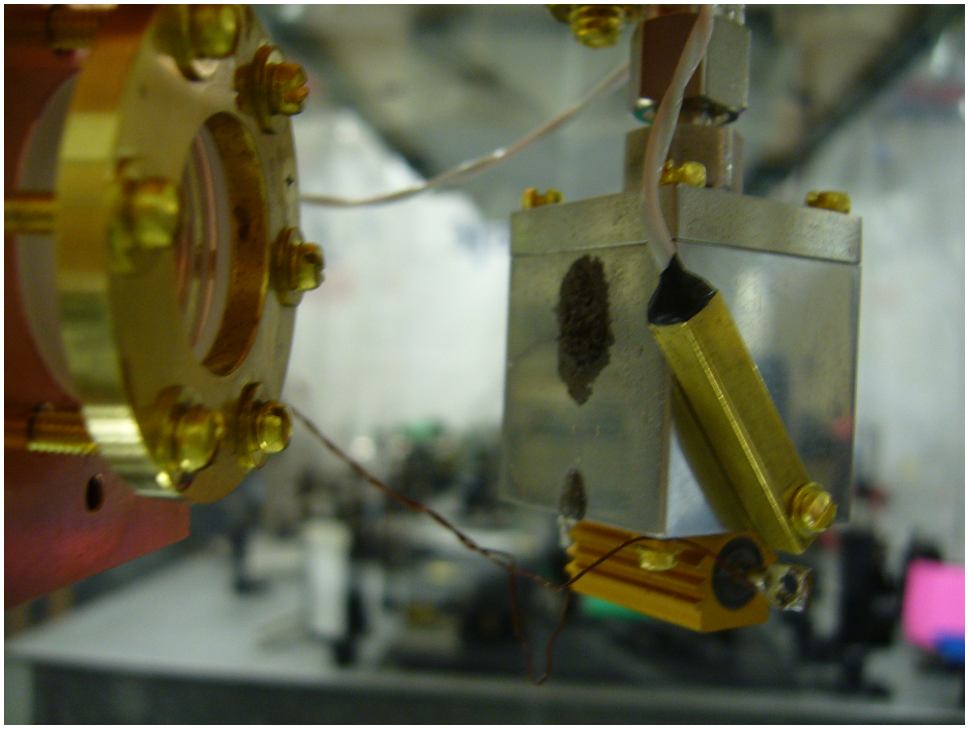


Figure 3.3: Image of the oven with its nozzle being clogged by oxidized rubidium. The resistor at the bottom of the oven was later moved to the nozzle-side after an oven redesign.

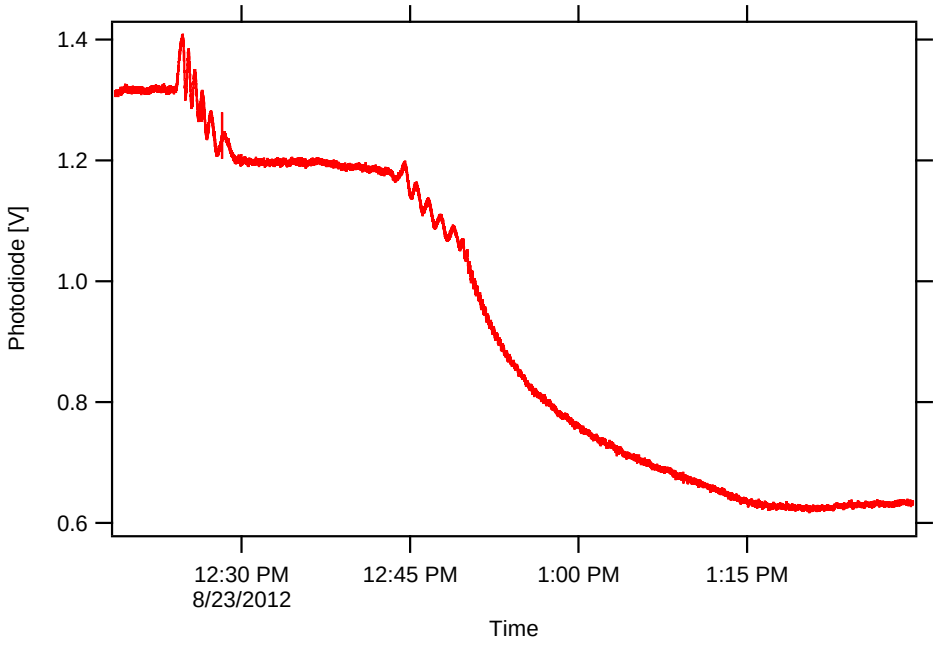


Figure 3.4: Photodiode signal during argon deposition at 4.8 K.

loaded rubidium into the oven.

In the second trial too we did not heat sapphire plate, but heated only the oven to unclog the nozzle. The sapphire window was maintained at temperature of about 4.8 K. At 269 K of oven temperature we observed a flow of argon detected by oscillating signal in the photodiode. But this time also the crystal was snowy or cloudy. The pressure of argon in oven was between 4 to 8 torr, and the chamber was maintained at 1×10^{-5} torr. The oven temperature was mostly at 283 K during the deposit. During the third trial we maintained sapphire window at 4.8 K, oven temperatue at about 283 K, oven pressure between 6 to 8 torr, and again the deposit formed was cloudy. Figure 3.4 shows the photodiode signal during the third trial of crystal growth. Although we can see some oscillations in the photodiode signal due to thin film interference, the signal is falling down rapidly indicating that the deposition did not lead to a transparent matrix.

We wanted the argon crystals to be as transparent as possible, but we observed

that at the temperature below 5 K the argon deposit is snowy. However, it is known that higher temperatures (20-25 K) are favorable for growing high-optical quality crystals due to better annealing [22, 57]. Therefore, we grew other crystals at 20 K as described in the subsection 3.1.5 below.

3.1.5 Growing argon crystal at 20 K

During the fourth trial of argon deposition we maintained the temperature of the sapphire window at 20 K, with oven pressure between 7 to 8 torr, and oven temperature around 273 K. This time we observed growth of a clear and transparent argon crystal. As shown in figure 3.5, the oscillation fringes were nice, and we could count 123 fringes before signal to noise ratio became too small. Figure 3.6 presents the zoomed-in view of the signal from the beginning part of the deposition. The rate of growth was 2.3 fringes per minute which is 0.6 μm per minute. With that rate we calculated the total thickness of the crystal formed to be 48 μm. The long-term oscillation with period of about 24 minutes is due to photodetector drifting over time.

Out of the next eleven trials of creating argon crystals only one resulted into a good crystal. For all of these trials, we did not warm-up the cryostat up to room temperature after deposit. We would melt the crystals at 50 K, and then cool down the cryostat. But later we found that warming the cryostat up to room temperature as described in subsection 3.1.3, and then cooling back to 20 K resulted in good crystals more often.

3.2 Growing argon crystal doped with rubidium

Our goal is to isolate rubidium in an argon crystal in such a way that the rubidium atoms do not clump together and are separated from each other by argon atoms.

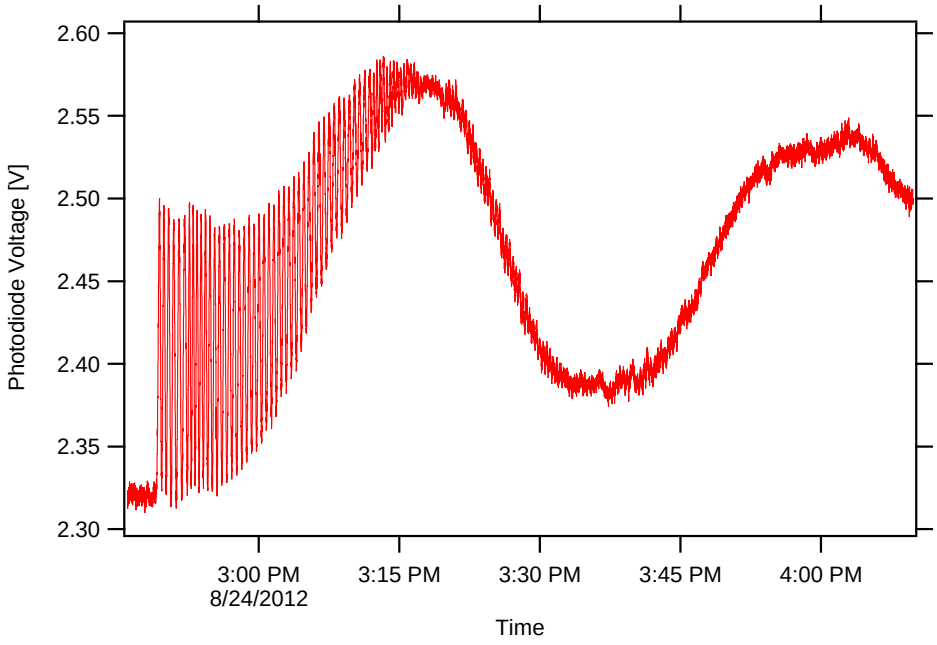


Figure 3.5: Photodiode signal during argon deposition at 20 K. The rapid oscillations are due to thin film interference. The long-term oscillations are due to the photodetector drift.

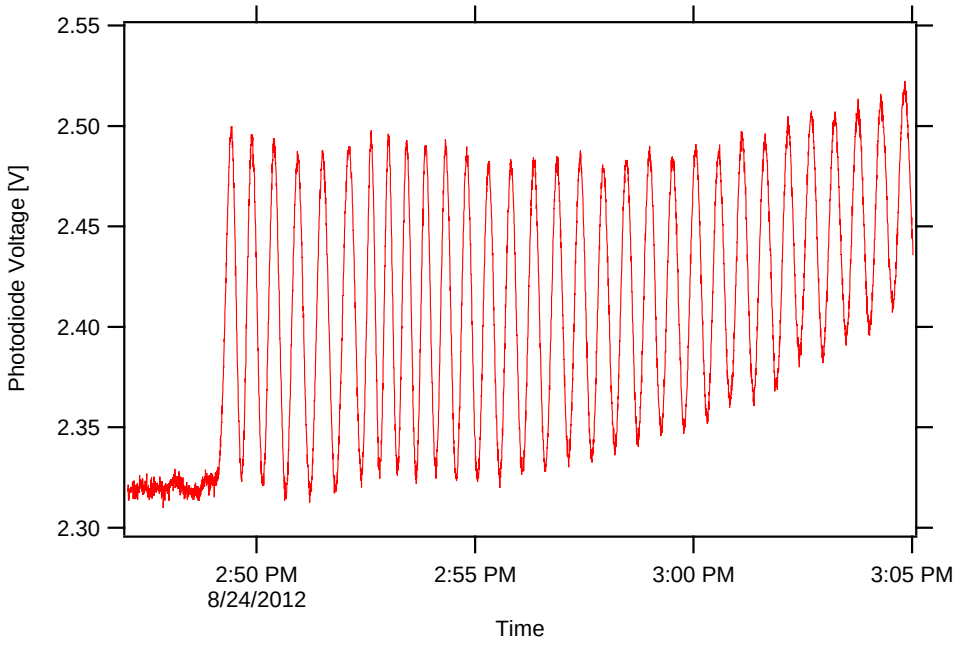


Figure 3.6: Zoom-in view of the initial part of the photodiode signal during argon deposition at 20 K.

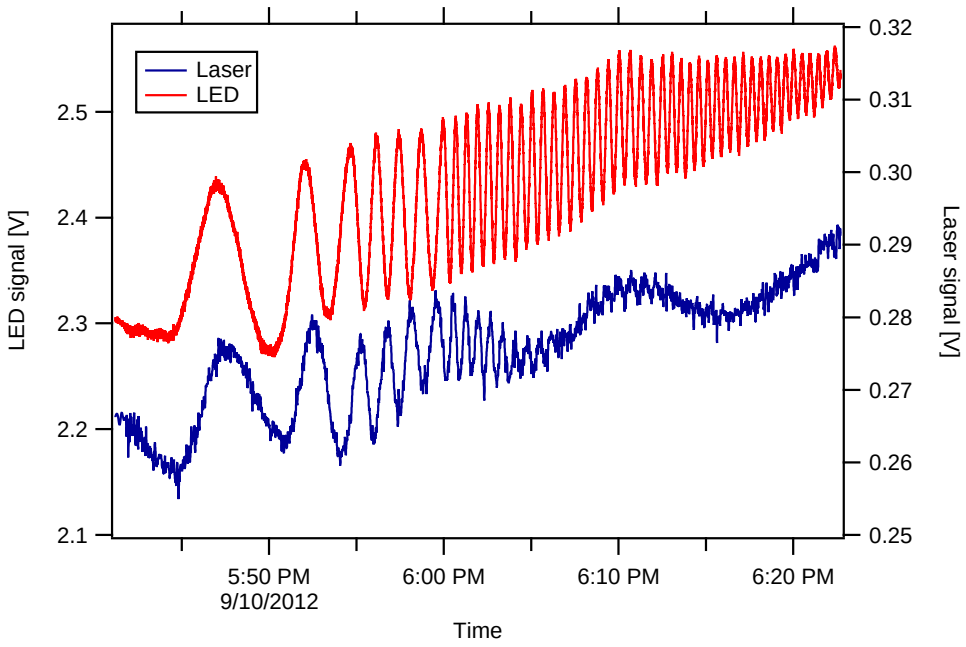


Figure 3.7: Signals during growth of the crystal, obtained using an LED at wavelength 635 nm and laser at wavelength 799 nm.

Our aim was to have the ratio of number of rubidium atoms to the number of argon atoms between 1:1000 to 1:10000. This was based on the experiences of Kupferman and Pipkin, who in 1960's isolated rubidium in argon matrix [2]. It is important to find out the right oven temperature to control the amount of rubidium atoms during deposition. Therefore, we grew many Ar-Rb crystals one after another under different oven temperatures each time. We used the vapor pressure curve as presented in appendix B to choose the possible right oven temperature. The greater the temperature the more rubidium we should get, as shown in the vapor-pressure curve. Thus we could control the amount of rubidium by controlling the oven temperature. Also, we could control the amount of argon by controlling its pressure in the oven.

We monitored the optical density of the crystal as a function of wavelength by using a miniature grating spectrometer from Ocean Optics [48]. The spectrometer has a wavelength range of 350-1000 nm, an entrance slit of 25 μm, and has an optical resolution of $\sim$2.0 nm [48]. We used a common incandescent lamp or a halogen lamp

as a light source for this spectrometer.

Our general strategy while creating Ar-Rb crystal was to first grow 5-10 microns crystal with argon only, then rapidly heat the oven to the appropriate temperature to add rubidium to the flow, and continue to grow argon with rubidium until we get an OD of about 1, and then turn off the oven heater to grow another 5-10 microns with argon only. Thus the middle part of Ar-Rb crystal would be sandwiched between argon-only layers.

Following this strategy we grew many crystals at different oven temperatures - starting from about 273 K to 431 K. Until we tried 431 K we did not see any significant change in OD, but when we grew a crystal at $\sim$431 K we observed signal in the spectrum. We grew this crystal with sapphire window temperature of 20 K, and oven pressure of $\sim$45 torr. The growth rate was $\sim$0.5 μm per minute and the growth duration was about 3 hours. From the oven pressure and temperature we calculate rubidium to argon ratio to be 1 to 8000. Figure 3.8 shows the absorption signal from which we calculate the rubidium density as 4×10^{18} cm^{-3}, using the method described in section 3.3. In the figure 3.8 we can observe a significant broadening to the blue and red sides of rubidium absorption lines 780 nm and 795 nm. We see that the broadening is significantly worse than that of the experiment by Kupferman and Pipkin. [2] (shown in figure 3.11). We lost this crystal due to power outage, the problem which is discussed in subsection 3.2.3.

After successive growths of crystals the sapphire window became too dirty to grow any more good crystals, and therefore we opened up the cryostat to clean the window. Then we grew more crystals varying the oven temperature each time. We wanted to find the right oven temperature at which we can achieve gradual increase of OD ensuring more or less homogeneous distribution of rubidium in the crystal. A typical signal of photodiode during a growth of Ar-Rb crystal is shown in figure

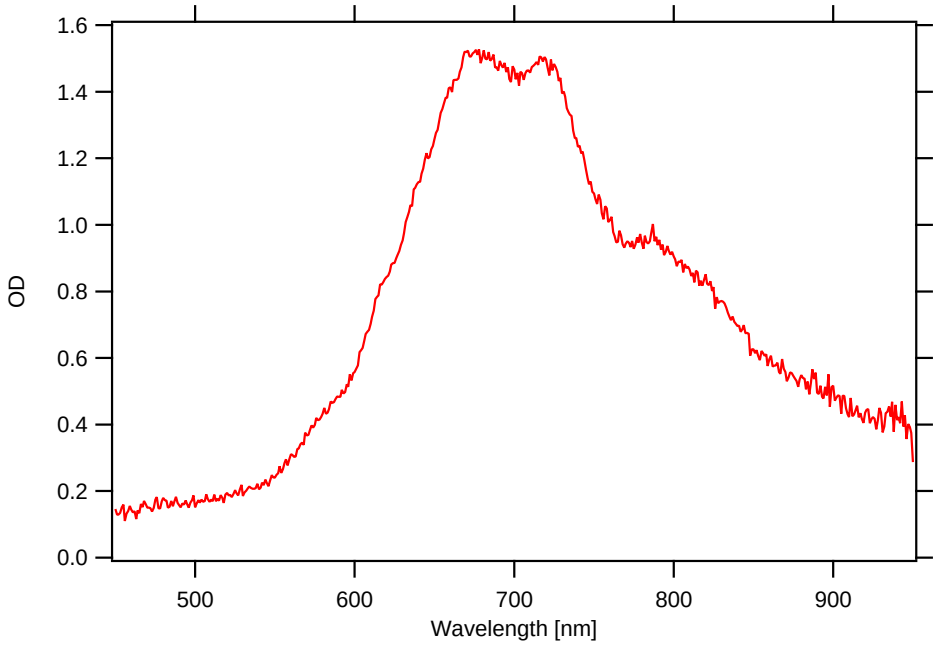


Figure 3.8: Absorption spectrum of rubidium in solid argon matrix at 20 K. We see significant broadening to the blue and red sides of Rb absorption lines 780 nm and 795 nm.

3.9. At about 315-320 K of oven temperature we observed a desired OD and rate of rubidium deposition. But still the OD signal was broad similar to that of figure 3.8, and we did not see the nicely defined peaks at wavelengths as described by Kupferman and Pipkin [2]. Then we decided to change the parameter temperature of sapphire window during the crystal growth.

Initially, we had the problem of nozzle being clogged due to oxidized rubidium, as can be seen in the figure 3.3. This was due to rubidium condensing in the nozzle as it was not heated properly by the heater which was attached to the oven base. In order to avoid this problem we redesigned the oven and moved the resistor heater from the base of the oven to the nozzle side. This helped to get a better flow of Ar-Rb than with previous design.

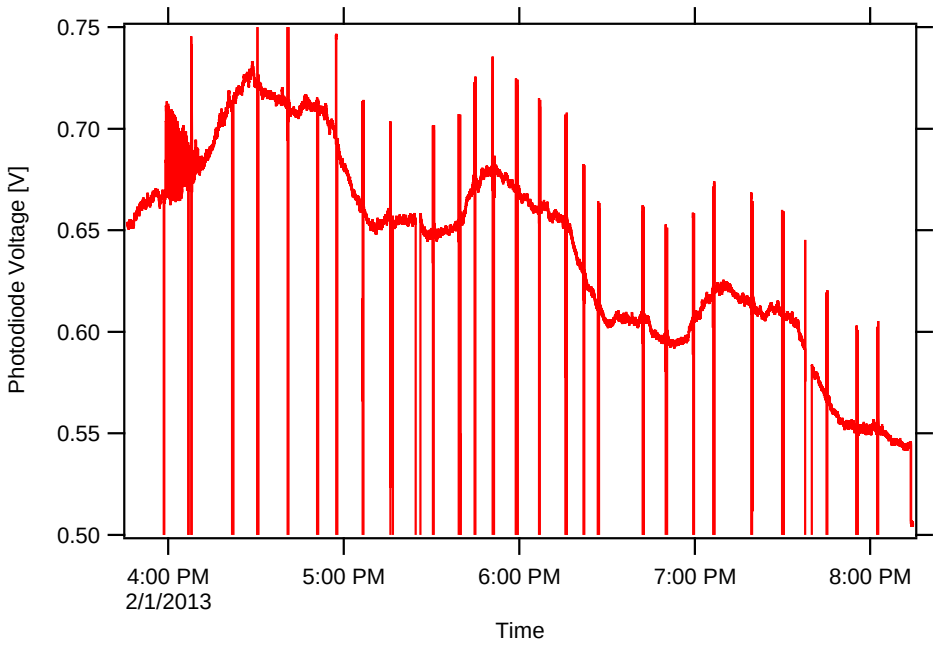


Figure 3.9: Photodiode signal during the growth of a typical crystal. The vertical lines are due to interruptions of LED light while taking spectrum using broadband light and spectrometer.

3.2.1 Growing Ar-Rb crystal at 15 K

We decided to grow crystals at 15 K to see if this helps to achieve spectrum with narrower lines. The downside of using 15 K could be worse annealing [22]. We used oven pressure of ∼45 torr, and oven temperature of ∼316 K. The duration of growth was ∼3.5 hours, and the growth rate was 1 μm per minute resulting into a ∼200 μm thick crystal. The rubidium density and the rubidium to argon ratio is calculated in section 3.3 below.

The crystal grown at this temperature (with parameters like oven temperature kept the same as before) gave a nice spectrum comparable to the spectrum of Kupferman and Pipkin [2]. Figure 3.10 presents the various spectra during and after the growth of the crystal at 15 K. We can see that during the growth of crystal the spectra are broad. But once we cooled the crystal below 10 K, nicely defined peaks appear around the wavelengths 799 nm, 785 nm, and 747 nm.

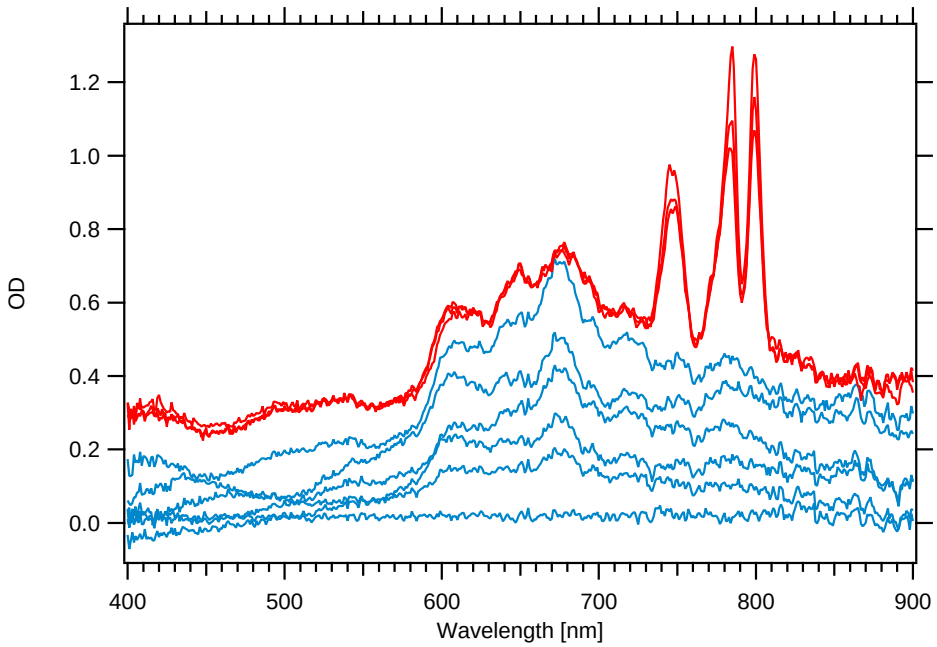


Figure 3.10: Spectra during and after the growth of Ar-Rb crystal. The spectra in blue are during the growth of crystal at 15 K. The spectra in red are obtained after cooling down the crystal below 10 K. The three sharp peaks are at wavelengths 799 nm, 785 nm, and 747.

Figure 3.11 presents the absorption spectrum from the paper by Kupferman and Pipkin, and figure 3.12 presents the spectrum from our crystal. The spectrum obtained by Kupferman and Pipkin [2], shows that there are 7 rubidium lines in argon between 650 nm and 800 nm, of which three are red-shifted and four are blue-shifted. The wavelengths at the three lines on the red side of our spectrum match closely with those of Kupferman and Pipkin, and they are narrower with higher OD. But the other four peaks on the blue side of our spectrum are broader than those in their spectrum. But we are interested in optical pumping of rubidium at the wavelength around 799 nm (because at this wavelength Kupferman and Pipkin observed the highest difference of absorptions of σ^+ and σ^- polarized light, as shown in figure 3.11). Therefore we preserved this crystal to work on optical pumping as described in chapter 4. As with most of the crystals, this crystal also cracked while cooling down gradually from 15 K to 3.2 K in 1 hour, but fortunately it did not fall off the sapphire window.

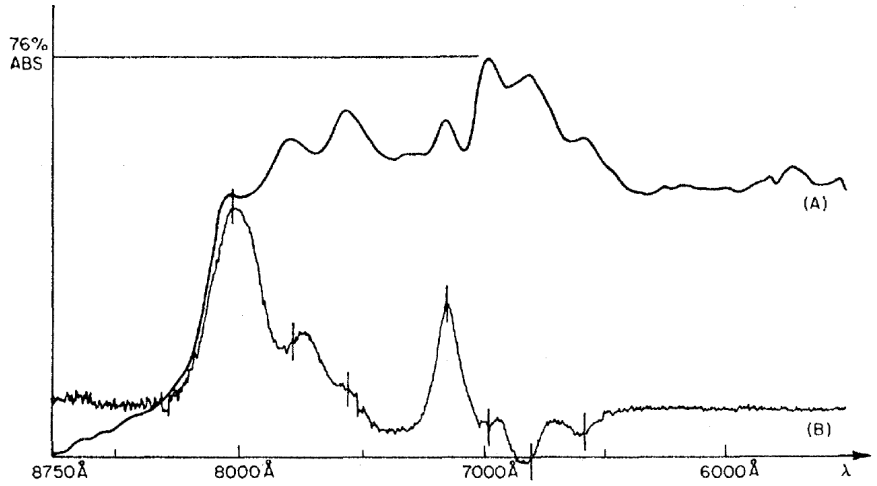


Figure 3.11: Figure obtained from the paper of Kupferman and Pipkin [2]. The upper curve (A) is the absorption signal, and the lower curve (B) is the motorized circular polarizer signal: the difference of absorption of σ^+ and σ^- polarized light.

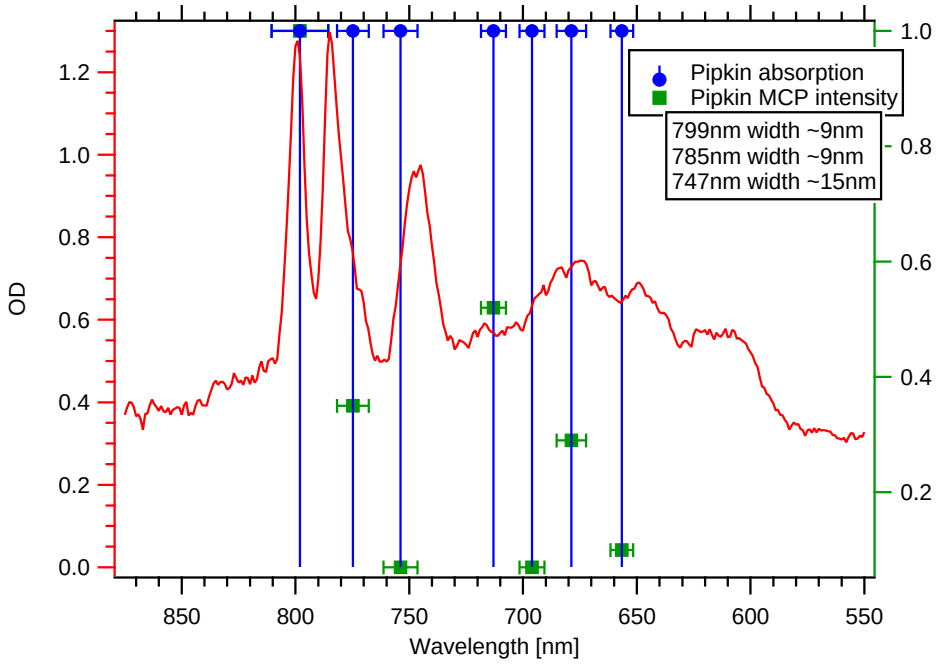


Figure 3.12: Spectrum of rubidium trapped in the argon crystal formed at 15 K in our laboratory. The widths of the lines at 799 nm and 785 nm are ∼9 nm at the OD of ∼0.9. The width of the line at 747 nm is ∼15 nm at the OD of ∼0.7. The blue circles indicate the absorption lines with the associated error bars from the absorption spectrum of Kupferman and Pipkin. The green squares indicate their motorized circular polarizer (MCP) relative intensities (right axis) at the different lines with associated error bars [2].

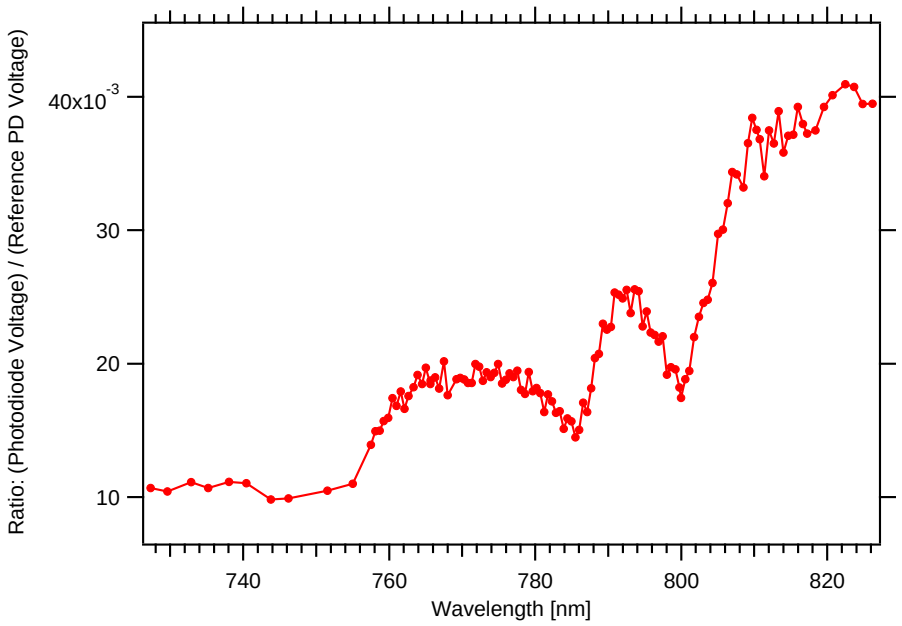


Figure 3.13: Spectrum using laser, obtained by changing the wavelength and briefly unblocking the laser at each wavelength for 2 s. Y-axis is the ratio of photodetector signal (with crystal) to the reference photodetector signal (without crystal). We can notice the absorption lines at 785 nm and 799 nm, and compare with the spectrum in the figure 3.12 taken with broadband light.

We subsequently took a spectrum using a laser (Ti:Sapph) by changing the laser wavelength manually. At first we did not observe defined spectrum lines as we saw in the spectrum using a lamp. As described in section 4.1 we later found out that the problem is "bleaching" of the crystal when the laser is on for a long time while taking the spectrum. Therefore, we obtained a spectrum by changing laser wavelength and briefly unblocking the laser at each wavelength for 2 s. This is presented in figure 3.13, which matches our spectrum using the lamp as shown in figure 3.12. The successful optical pumping experiment based on this particular crystal is described in chapter 4.

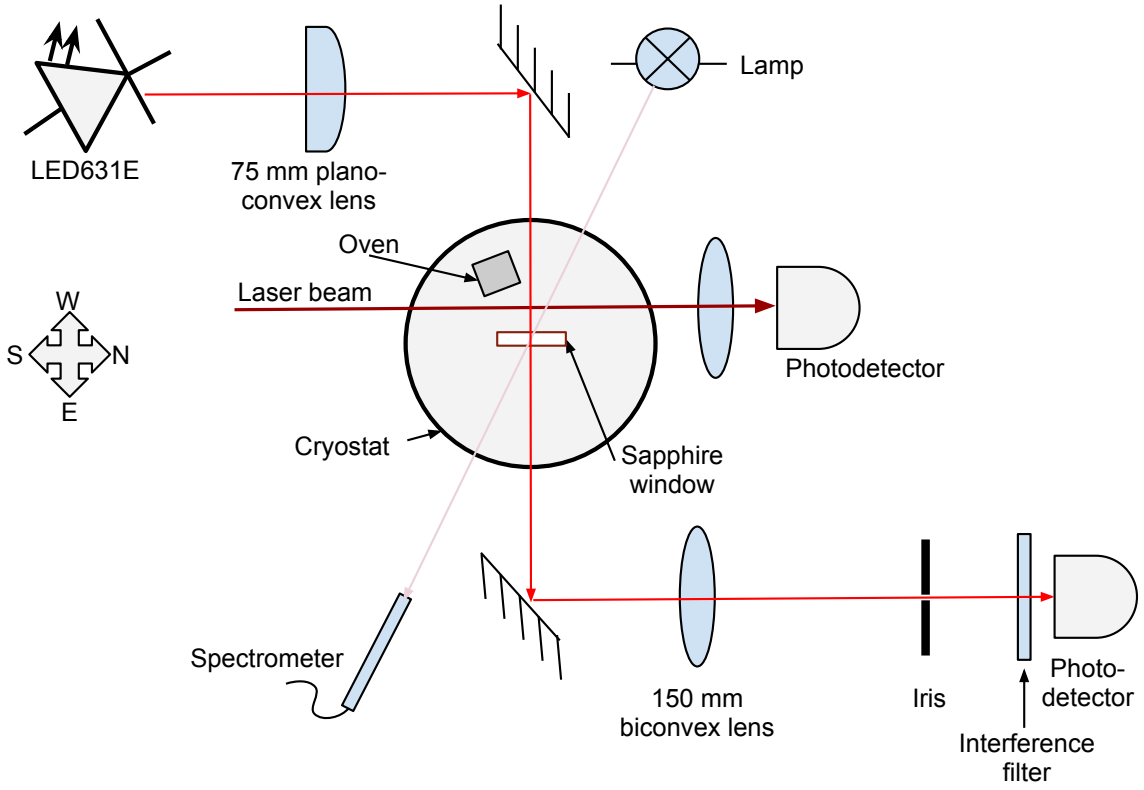


Figure 3.14: Schematic of the optics setup to monitor the growth of argon crystals doped with rubidium.

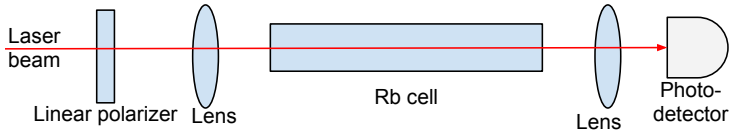


Figure 3.15: Rb Reference cell setup.

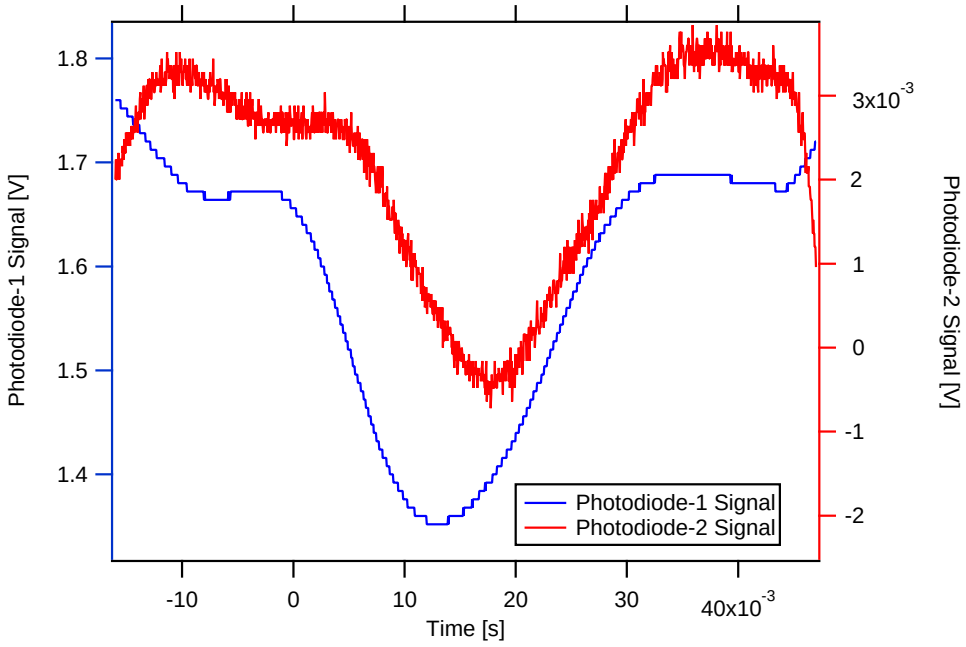


Figure 3.16: Photodiode signals, red showing detection of rubidium in the Ar-Rb flow out of oven, blue showing the rubidium signal from the reference Rb cell. Laser is scanned around the D_1 resonance line of rubidium.

3.2.2 Detecting rubidium in Ar-Rb gas beam

To monitor the rubidium flowing out of the oven we added more optics in addition to those shown in figure 3.1. The new optics setup is shown in figure 3.14. We pass a laser light in north-south direction (crossing the path of Ar-Rb flow but not at perpendicular). For reference we setup a Rb vapor cell as shown in figure 3.15. Since the density of rubidium is low in the flow, it is hard to detect rubidium signal. Nevertheless, we were able to detect rubidium with an OD of the order of 10^{-4}. Figure 3.16 shows the rubidium signal detected in a Rb reference cell and the signal detected in the Ar-Rb flow when the laser is scanned around the D_1 resonance line of rubidium. The shift seen in the Rb signal of the flow can be explained by the fact that the laser looking at the Ar-Rb flow is not perpendicular but at some angle with the direction of the flow.

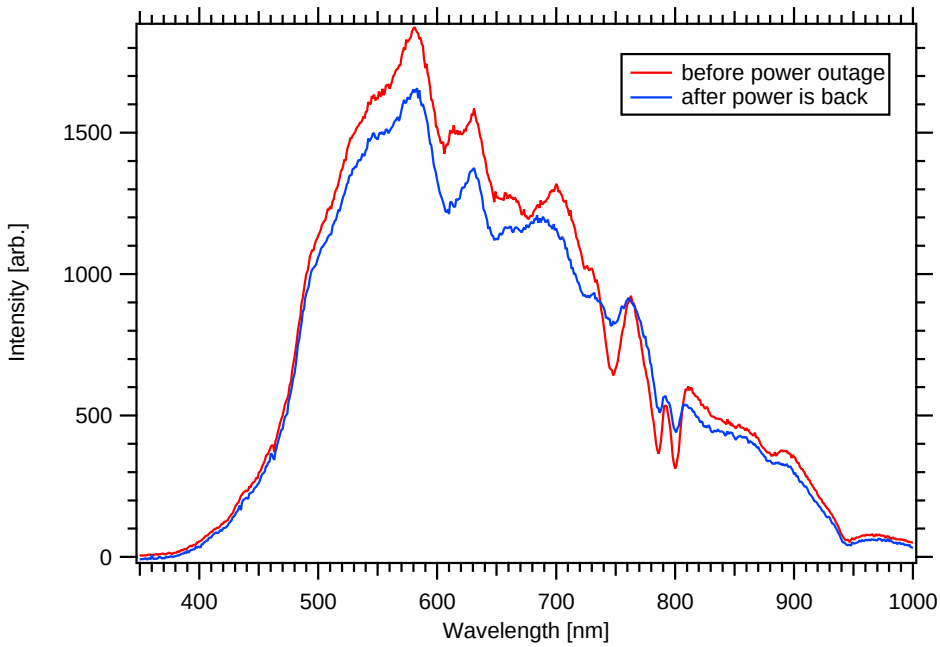


Figure 3.17: Spectrum of rubidium trapped in argon crystal before (in red) and after (in blue) power outage. Notice the change in intensity at the lines 747 nm, 785 nm, and 799 nm.

3.2.3 Power outage and its effect on spectrum

Unfortunately, before we could start optical pumping experiment in this good crystal, we had a power outage for about 45 minutes. This raised the temperature of sapphire window to slightly more than 17 K. When the power was back we took spectrum, and found that the absorption at red group of lines had decreased by a factor of 3. We surmise that it would reduce the signal to noise ratio in optical pumping experiment also by about a factor of 3. But still, as shown in figure 3.17, we observed a distinguishable absorption peak at around 799 nm, and therefore we decided to continue to work on this crystal for optical pumping. In the figure 3.17 we can see the change in spectrum caused by the power outage.

3.3 Rubidium density in the Ar-Rb crystal

In this section we calculate the density of rubidium atoms in the Ar-Rb crystal (deposited at 15 K) on which we perform optical pumping as presented in Chapter 4.

First we find the density by using the value of optical density (OD) obtained from figure 3.12. OD of a sample is related to density as

$$OD \approx \sigma \cdot \rho \cdot l = \sigma \cdot \rho_{areal} \tag{3.2}$$

where σ is the absorption cross-section, ρ is the volume density and ρ_{areal} is the areal density of rubidium in the crystal, and l is the thickness of the crystal. σ is given by [31]

$$\sigma \approx \frac{\lambda^2}{2\pi} \cdot \frac{1}{2\pi\Delta\nu} \cdot \frac{1}{\tau} \tag{3.3}$$

where λ is the wavelength, $\Delta\nu$ is the full width at half maximum, and τ is the excited state lifetime.

As a first way of calculating the rubidium density we use the OD of the solid Ar-Rb crystal. First, we calculate the total rubidium density by considering the entire absorption spectrum presented in figure 3.12. At $\lambda = 670 \times 10^{-9}$ m, we have OD of 0.4 and $\Delta\nu \approx 1.3 \times 10^{14}$ Hz. Then with $\tau \approx 27 \times 10^{-9}$ s [58], we get $\sigma = 9.0 \times 10^{-21}$ m^2. Then substituting this value of σ to equation 3.2, and with $l = 200 \times 10^{-6}$ m and OD = 0.4, we get the volume density, $\rho = 2.2 \times 10^{23}$ m^{-3} $= 2.2 \times 10^{17}$ cm^{-3}, and the areal density, $\rho_{areal} = \rho \cdot l = 4.4 \times 10^{19}$ m^{-2} $= 4.4 \times 10^{15}$ cm^{-2}.

Next, we consider the subset of rubidium atoms which interact with 799 nm laser. We find that at $\lambda = 799 \times 10^{-9}$ m, OD is about 1.0 and $\Delta\nu \approx 4.7 \times 10^{12}$ Hz. Then with $\tau \approx 27 \times 10^{-9}$ s [58], we get $\sigma = 1.3 \times 10^{-19}$ m^2. Then substituting this value of σ to the equation 3.2, and with $l = 200 \times 10^{-6}$ m and OD = 1.0, we get the volume density, $\rho = 3.8 \times 10^{22}$ m^{-3} $= 3.8 \times 10^{16}$ cm^{-3}, and the areal density, $\rho_{areal} = \rho \cdot l = 7.7 \times 10^{18}$ m^{-2} $= 7.7 \times 10^{14}$ cm^{-2}.

As a second way of calculating the rubidium density we can use the OD of the gas beam of Ar-Rb mixture during deposition, as described in section 3.2.2. The OD was $6 \pm 2 \times 10^{-5}$, and the full width at half maximum at $\lambda = 795 \times 10^{-9}$ m is $\Delta\nu \approx 1 \times 10^{9}$ Hz. Then from equation 3.3 we find $\sigma = 5.9 \times 10^{-16}$ m^2. Substituting this value of σ to the equation 3.2, and assuming $l = 2$", we find density of rubidium in the gas beam as $n_{gas} = 2 \times 10^{12}$ m^{-3}. The velocity of rubidium atoms at 316 K is calculated to be $v = 3 \times 10^{2}$ ms^{-1}. Then the rubidium deposition rate per unit area per unit time is $n_{gas} \cdot v = 6 \times 10^{14}$ m^{-2}s^{-1}. This gives the areal density of rubidium in the solid crystal formed in the duration $t = 3.5$ hours as $\rho_{areal} = n_{gas} \cdot v \cdot t = 7.6 \times 10^{18}$ m^{-2} $= 7.6 \times 10^{14}$ cm^{-2}. Thus the rubidium density in the solid crystal is $\rho = \frac{\rho_{areal}}{200 \times 10^{-4}} = 3.8 \times 10^{22}$ m^{-3} $= 3.8 \times 10^{16}$ cm^{-3}. We can see that this density of rubidium is an order of magnitude smaller than the total rubidium density calculated from the absorption spectrum of the solid crystal. This discrepancy is not understood.

As a third way of calculating the rubidium density, we use the pressure of argon and rubidium inside the oven. As described in 3.2.1 we used argon pressure $P_{Ar} \approx 45$ torr in the oven, and oven temperature of $\sim$316 K. Using rubidium vapor pressure curve in appendix B, we find that the pressure of rubidium at 316 K is $P_{Rb} \approx 3 \times 10^{-6}$ torr. The ratio of argon pressure to rubidium pressure gives the ratio of number of argon atoms to rubidium atoms, which is 1.3×10^{7}. With solid argon density $\sim$1.7 g cm^{-3} and with argon atomic weight of 40, we find that the total number of argon atoms in 1 cm^3 is 2.6×10^{22}. Then the density of rubidium in the crystal $= \frac{2.6 \times 10^{22}}{1.3 \times 10^{7}}$ cm^{-3} $= 2 \times 10^{15}$ cm^{-3} $= 2 \times 10^{21}$ m^{-3}. This density is significantly lower than the density calculated using the other two ways above. Based on this discrepancy we guess that the actual temperature of the oven during deposition could be higher than 316 K, and that the resistance detector [39] is not showing an accurate oven temperature.

Chapter 4

Optical Pumping

In this chapter we present experimental results on optical manipulation of rubidium atoms trapped in a solid argon crystal. We first discuss the problem encountered during the attempt of optical pumping and then its solution. Then we discuss the optics setup and the results obtained on the optical pumping of spin of rubidium atoms.

4.1 Problem of "bleaching" and its solution

After successful creation of rubidium-doped argon crystal (section 3.2.1) we attempted to take a spectrum using a laser. This is presented in figure 4.1. We had expected the spectrum to be similar to that obtained using the broadband light source as presented in figure 3.12. However, we did not observe the narrow spectral lines seen in the white-light spectrum.

We found that when laser light is used its transmission through the Ar-Rb crystal increases gradually over time. Using a photodetector we observed that the signal increases over time and then levels off becoming asymptotic after about 10 minutes. This observation is presented in figure 4.2. Fitting with the exponential function of

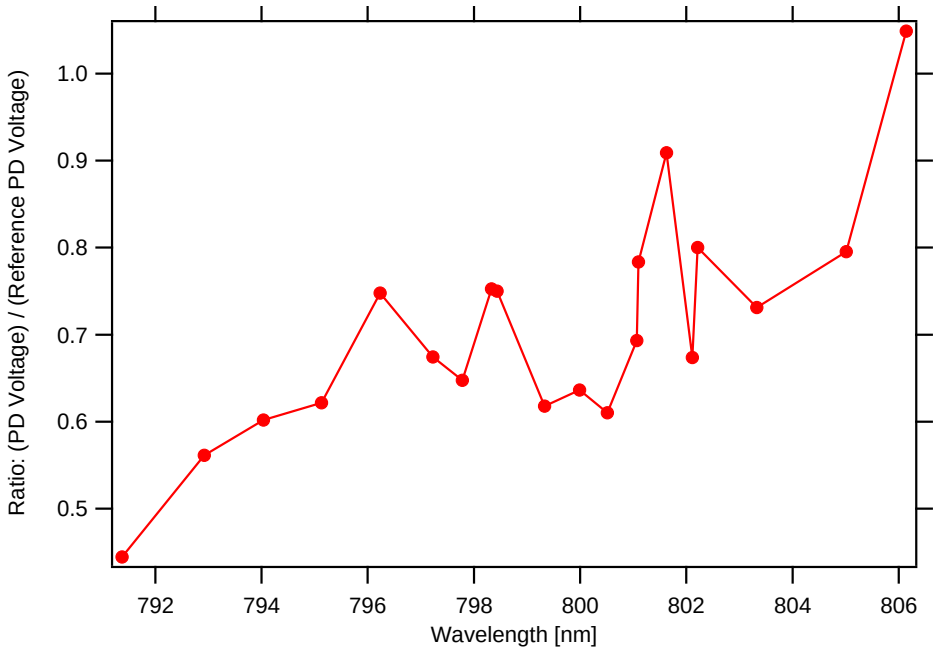


Figure 4.1: Spectrum using laser, obtained by changing wavelength manually. We do not see a defined line around 799 nm as we observed with broadband light source (compare with figure 3.12 of chapter 3). Laser power and diameter used were 13 μW and $\sim$1 cm respectively.

the form

$$y = y_0 + Ae^{-(x-x_0)/\tau}, \tag{4.1}$$

we can find the value of time constant τ, which is 9.2 minutes for data in figure 4.2. We call this phenomenon "bleaching", and this explains why initially we were not able to observe a line at 799 nm in the spectrum obtained by using laser. The physical mechanism of bleaching is discussed in section 4.1.1. As expected, the value of τ decreases exponentially with laser power, as presented in figure 4.5. The lower the laser power the longer is the time taken for bleaching. If the laser is left blocked for more than 30 minutes, when it is unblocked we could see the gradual bleaching again. This suggests that "unbleaching" occurs during the period when laser is off.

Therefore, in order to take spectrum with laser with no significant effect of bleaching, we blocked laser for most of the time and briefly unblocked it at each wavelength

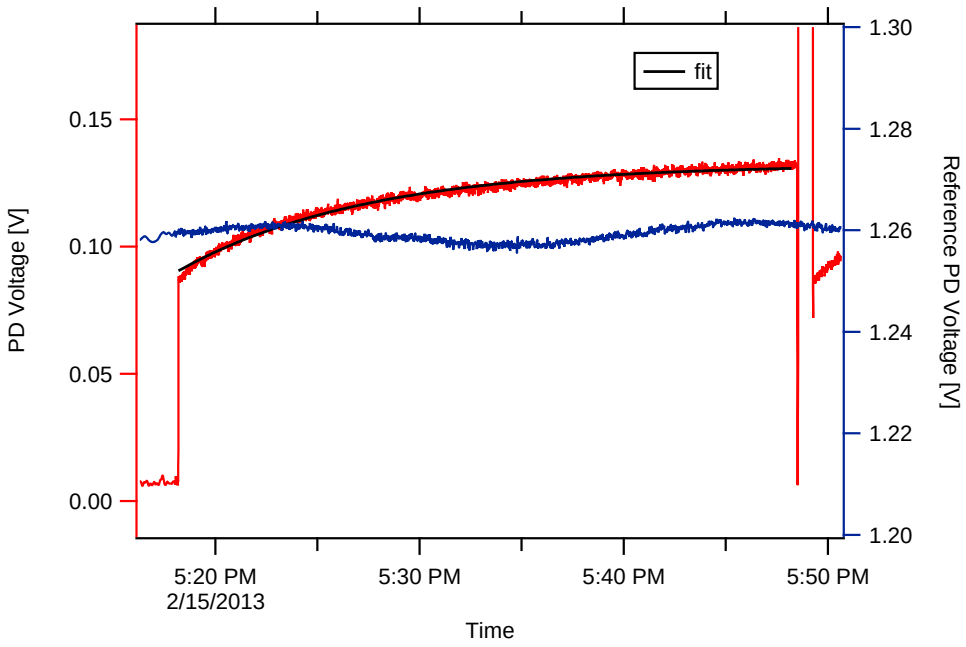


Figure 4.2: The phenomenon of "bleaching" observed when laser is passed through the Ar-Rb crystal. Laser power and diameter were 0.2μW and 1 cm respectively. The blue curve is the reference photodetector signal, and the red curve is the photodetector signal showing bleaching. The black curve is an exponential fit with function 4.1 and the value of τ extracted is 554.73 ± 1.42 s.

for 2 s. Then we obtained a spectrum as shown in figure 3.13. This matches with the spectrum using the lamp as shown in figure 3.12.

4.1.1 Lamp as an unbleaching light source

We later found out that the bleaching effect can be reduced significantly by shining white light from a lamp. As shown in figure 4.6, when white light is used, even if the laser is left on, the Ar-Rb crystal unbleaches rapidly. In the figure we can see the transmission decreases first rapidly and then slowly to level off. Thus continuous shinning of a broadband light on the Ar-Rb crystal helped to mitigate the effect of bleaching.

We observed that, as shown in figure 4.4, for any given laser power the transmission is lowest when we let the crystal unbleach completely by applying lamp light for about

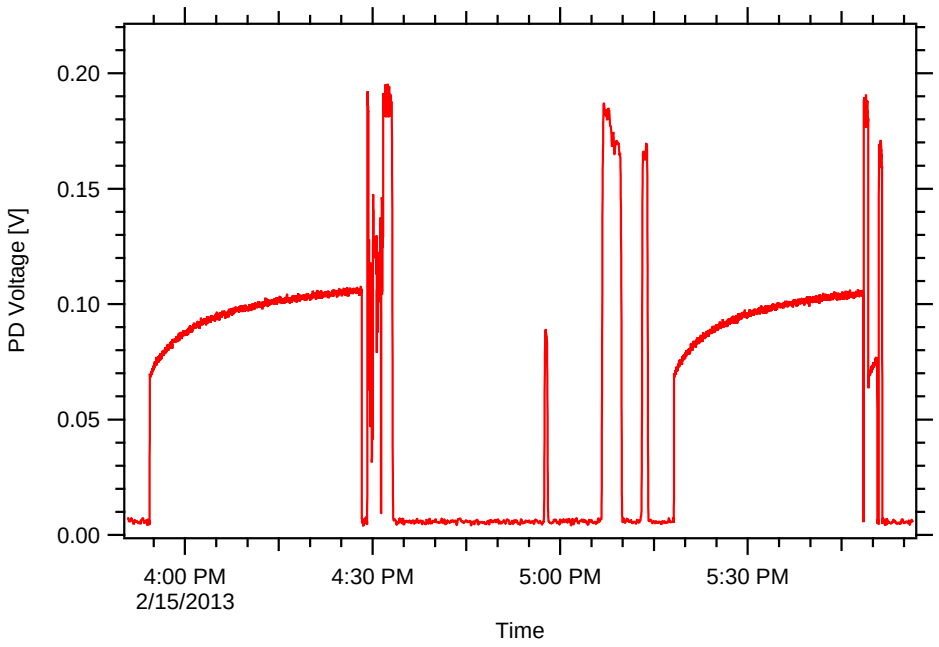


Figure 4.3: The curves from 3:55 PM to about 4:30 PM and from 5:20 PM to 5:50 PM show the rise in photodetector signal due to bleaching. Between 4:30 PM and 5:20 PM laser is left off when "unbleaching" occurs. The vertical lines are due to leakage of light from the lamp to the photodetectors.

10 minutes in absence of laser. When both laser and lamp are left on, we see that the transmission is still lower than that without lamp.

Figure 4.7 shows spectra with the laser on and off. When the laser is off and only the lamp is on, we see the absorption lines at 799 nm, 785 nm and 747 nm. But when the laser is turned on, we see the absorption decreases significantly around these three wavelengths. The sharp peak at 799 nm is due to the laser. Since the bleaching due to laser at 799 nm is effective not only at the line 799 nm but other lines as well, we conclude that we observe homogeneous broadening of these three red lines. Interestingly, we also see that when absorption decreases in the red lines it increases in the range of wavelengths 660 nm and 730 nm. This suggests that there are at least two long-lived trapping sites of rubidium in the argon crystal, and that population can be transferred optically from one site to the other. Therefore, if a light source with wavelengths in the range of 660 nm and 730 nm is used the red lines

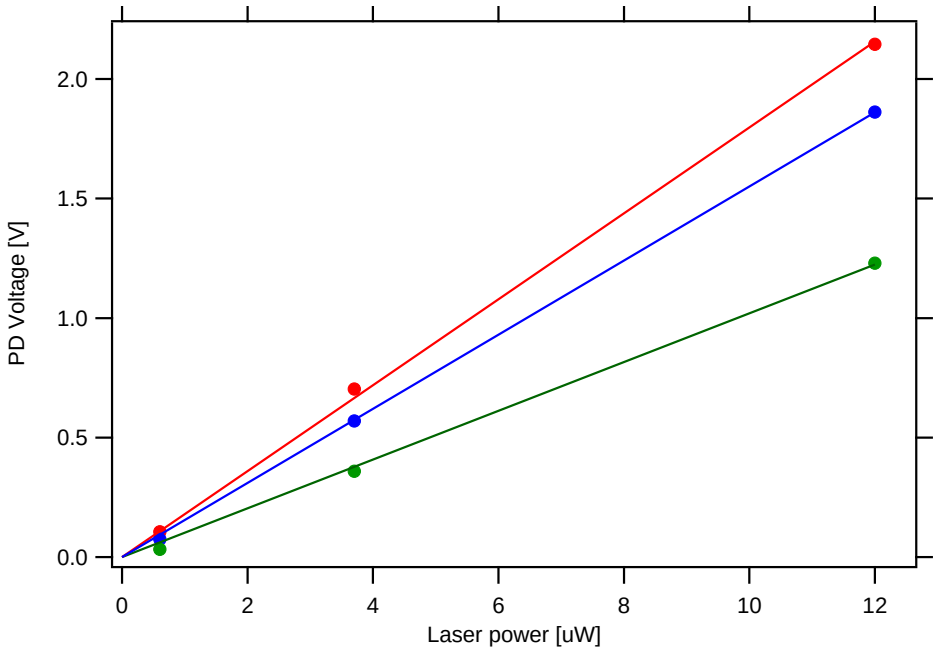


Figure 4.4: Effect of lamp upon bleaching due to laser at different powers. The red markers represent the maximum photodetector voltage obtained due to bleaching when only laser is on. The blue markers represent the same when both laser and lamp are on. The green markers denote the photodetector voltages at the beginning of bleaching (after a complete unbleaching due to a lamp).

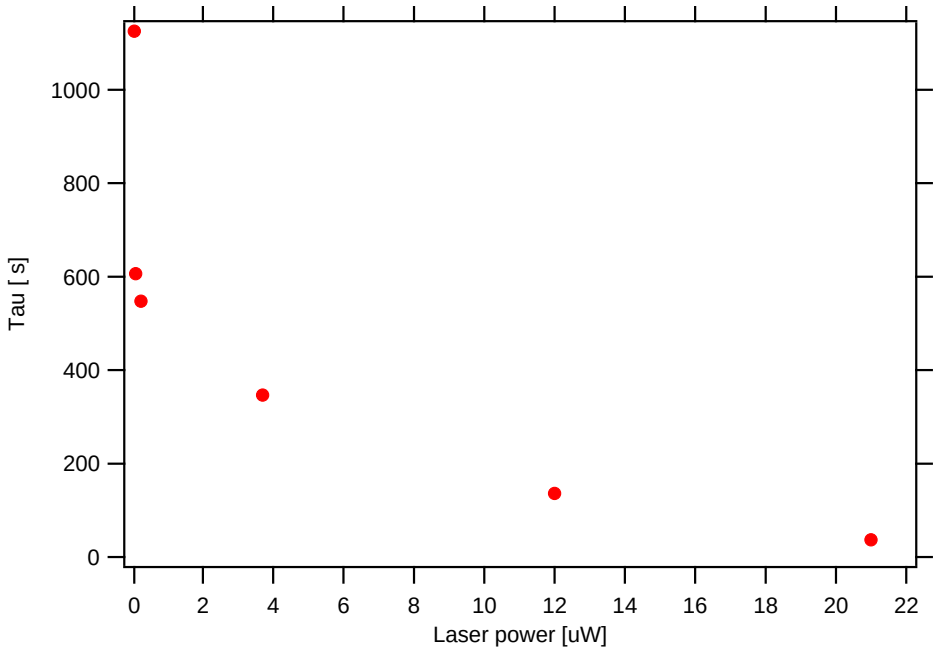


Figure 4.5: Dependence of τ of exponential fit (equation 4.1) on laser power during bleaching. Laser diameter was $\sim$1 cm at the crystal.

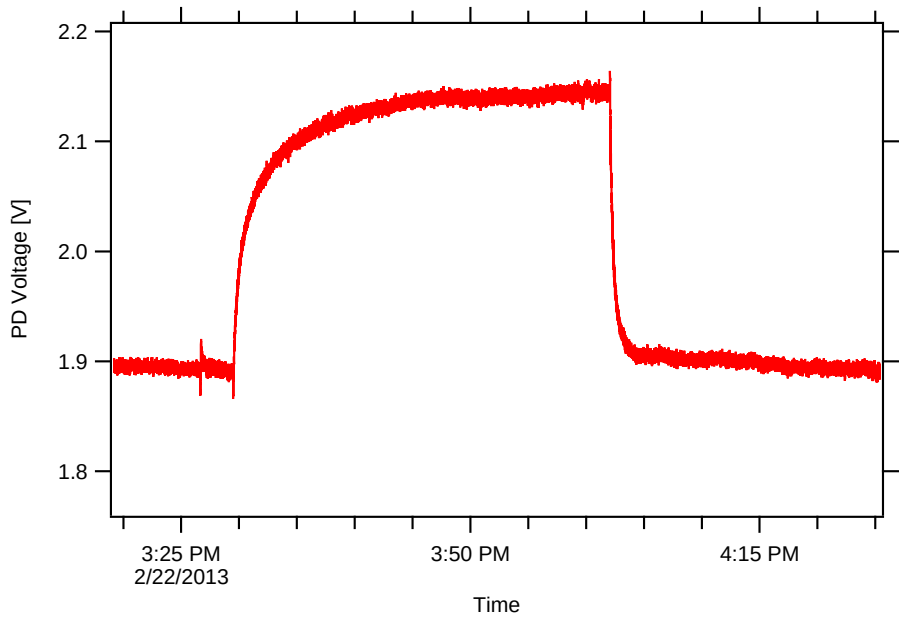


Figure 4.6: Unbleaching effect of white light using a lamp. Before 3:30 PM both laser and lamp are on. After 3:30 PM lamp is off and only laser is on so that the photodetector voltage increases rapidly due to bleaching. After 4:00 PM lamp is on again and the photodetector voltage decreases rapidly and reaches the level that was before 3:30 PM.

can be repopulated and the effect of bleaching could be mitigated to a greater extent. This also explains the physical mechanism of bleaching: the bleaching occurs when the laser light transfers population optically from the trapping site associated with red group of lines to the trapping site associated with the range of wavelengths 660 to 730 nm.

4.1.2 LED as an unbleaching light source

Based on the logic presented in the last paragraph of subsection 4.1.1, we used a high power single chip LED (H2A1-H690) [59] having a peak wavelength of 690 nm (covering the wavelengths 660 nm to 730 nm) and optical output power of 60 mW. Figure 4.8 shows the spectrum with the LED. We see a sharp peak at 799 nm (at which the laser is tuned) and the absorption at other wavelengths (785 nm and 747

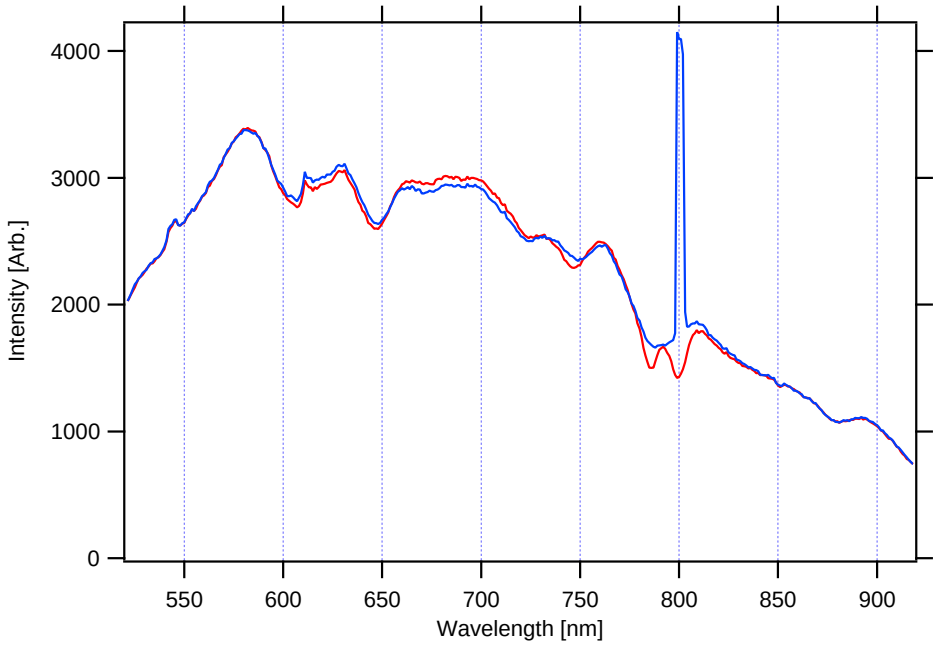


Figure 4.7: Spectrum of Rb in Ar crystal obtained using white-light source. The red curve is with white light only, and without laser. The blue curve is with both white light and laser. The blue peak at 800 nm is due to laser.

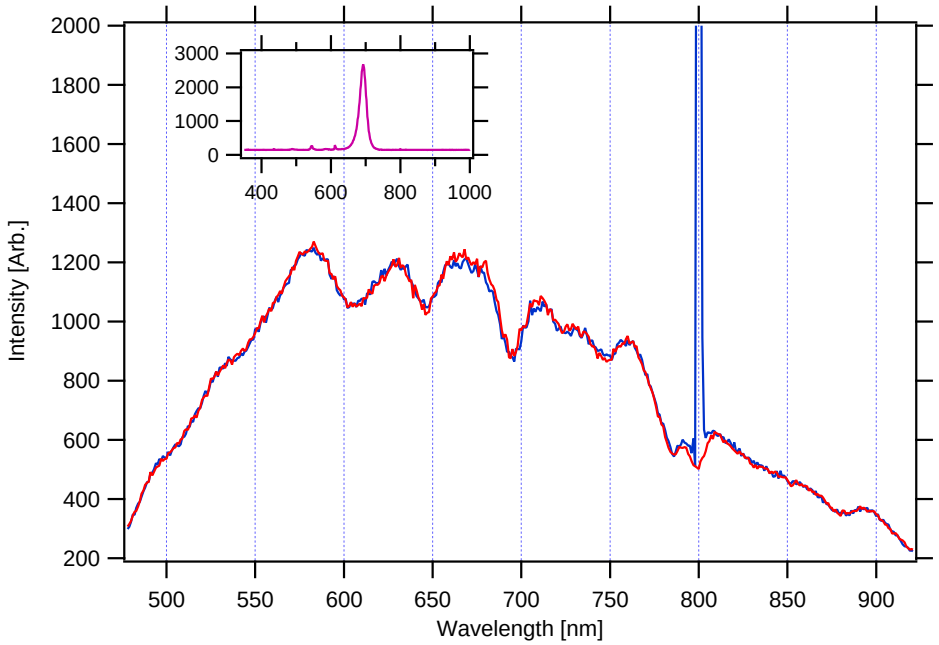


Figure 4.8: Spectrum of Rb in Ar crystal obtained using LED (with peak wavelength of 690 nm as shown in the inset). The red curve is with white light only. The blue curve is with white light and laser. The blue peak at 800 nm is due to laser. The positions of the light source and the spectrometer is different from those of figure 4.7.

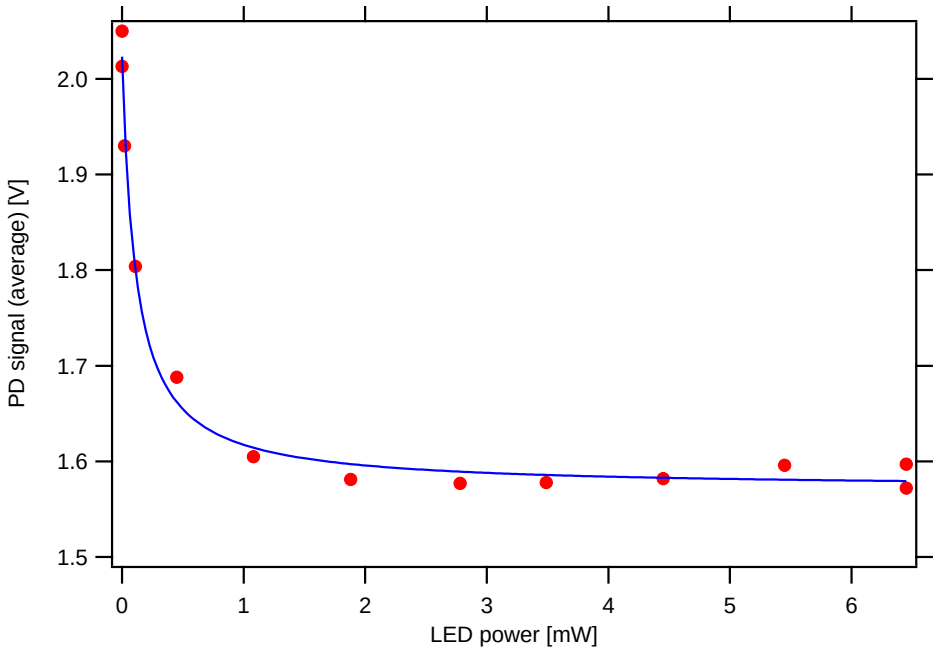


Figure 4.9: Dependence of bleaching effect on LED power. The markers are data, and the curve is the fit to the exponential function (equation 4.1). The laser power and diameter were 2.7 $\sim$ μm and was $\sim$0.4 cm (at the crystal). A 3" lens was used to focus the LED beam onto the crystal.

nm and in the range of wavelengths 660 nm to 730 nm) has not altered much.

We observed that the photodiode signal decreases with LED power. This has been presented in figure 4.9 which suggested that we need not use the maximum power of LED, and we could use as low as 1 mW to 2 mW in order to have effective unbleaching effect.

4.2 Optical Pumping

We use the pump-probe method to optically pump atoms and detect spin polarization and its relaxation. This is enabled by the unbleaching effect of a broadband light source (LED or lamp). By adjusting the intensity of light source for unbleaching, and by adjusting the duration of pump laser and the power of both pump and probe laser, we can control the bleaching effect and achieve optical pumping. For example, when

probe beam power is ∼0.5 μW (beam waist: ∼0.4 cm) and pump beam power is ∼2 mW (beam waist: ∼0.5 cm) we used LED power of ∼2 mW in order to minimize bleaching. Typical pump beam powers are from 1 to 3 mW, and the intensity of the beam on the crystal ranges from 1.6 to 4.7 mW/cm^2.

4.2.1 Measurement of relaxation time T_1

A schematic of the optics setup for the measurement of the relaxation time T_1 is presented in figure 4.10. Figure 4.11 shows the side view of the setup in simplified form. Using a wedge we split a laser beam into three beams. The high power beam is used as a pump beam, one of the low power beams is guided to a Coherent Wavemaster laser wavelength meter [60], and the other low power beam is used as a probe beam. By proper combination of angles of a half-wave plate and a quarter-wave plate we make the pump beam as circularly polarized (σ^+) as possible. We measured the σ^+ circular polarization of the pump beam to be 99%. Using a linear polarizer we make sure that the probe beam is linearly polarized. The probe and pump beam intensity at the Ar-Rb crystal are ∼0.01 mW/cm^2 and ∼2 mW/cm^2 respectively.

The pump and probe beams are guided and passed through the Ar-Rb crystal formed on the sapphire window. We overlap the pump and probe beam as much as possible with the help of CCD cameras. The typical angle between the beams is about 3°. On the other side of the cryostat the pump beam is blocked, while the probe beam is continued through a half-wave plate and a quarter-wave plate and then through a Wollaston prism. The Wollaston prism splits the probe beam into two orthogonal linearly polarized beams which are guided to two different photodetectors. By observing the photodetector signals of these two beams we can know the differential absorption of σ^+ and σ^- components of the probe beam.

The sapphire window used as the substrate is birefringent. In order to ensure this

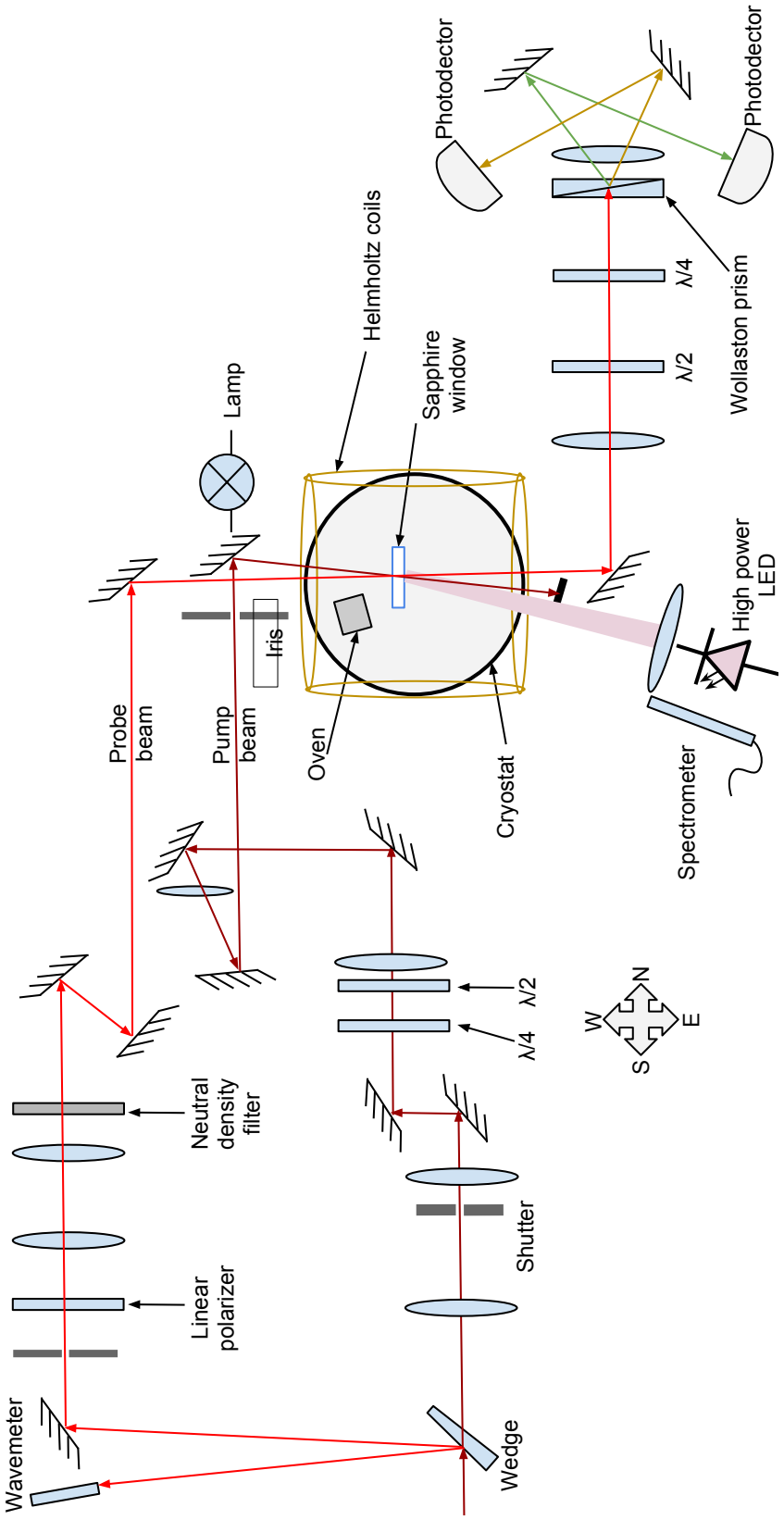


Figure 4.10: Schematic of the optics setup for the optical pumping experiment (top-bottom view).

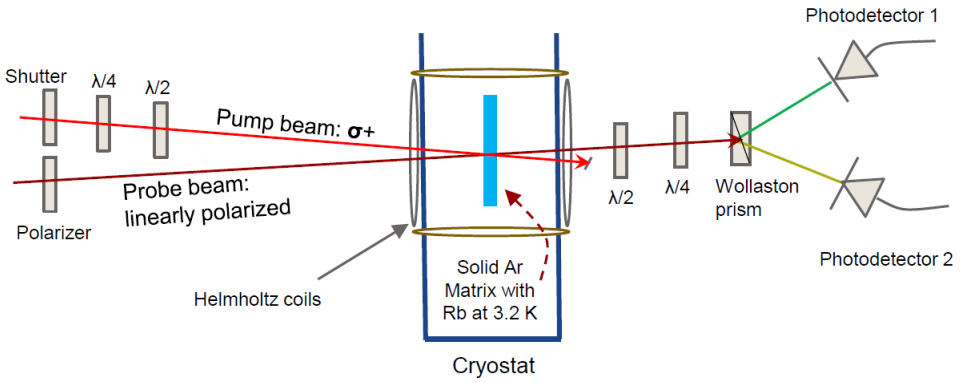


Figure 4.11: Simplified schematic of the optics setup for the optical pumping experiment (side view).

does not cause undue effect in the detection of differential absorption of σ^+ and σ^- components of the probe beam we had to carefully set up the wave-plates before the Wollaston prism. First, we changed the probe beam going into the cryostat to σ^+ by using a quarter-wave plate. Then, we aligned the quarter-wave plate and the half-wave plate in front of the Wollaston prism, and adjusted their angles to get almost all power to one of the detectors. Finally we changed the probe beam back to a linearly polarized light.

In order to optically pump and observe the relaxation of the polarization of rubidium ensemble we need to apply a uniform longitudinal magnetic field by using three pairs of Helmholtz coils in east-west, north-south and up-down direction. The coils are made of copper wires and their details are presented in table 2.1 of chapter 2.

To turn the pump beam on and off a shutter [61], having an aperture of 3 mm, is placed in its path as shown in figure 4.10. It has a minimum exposure time of ~ 4 ms, and has a rise time of 100 μs with a tightly focused beam. We control the pulse duration by controlling the opening and closing of this shutter by using transistor-transistor logic (TTL) signal sent to a shutter driver. We use a Labview program [52] in which we can input arbitrary time steps and sequences. Figure 4.12 presents an

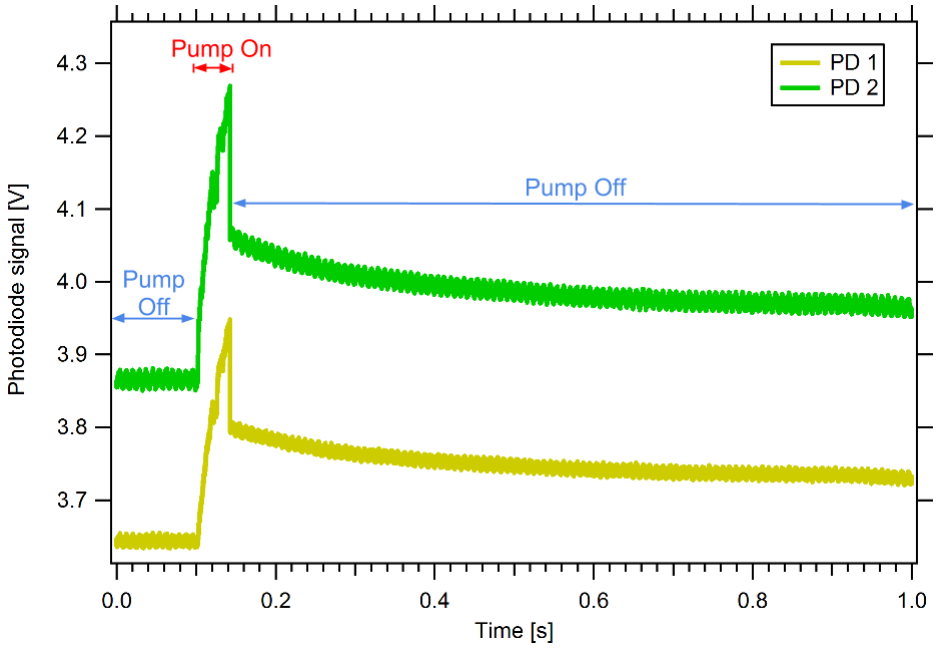


Figure 4.12: Raw data obtained during optical pumping experiment. The green curve (average of 4 observations) and golden curves (average of 2 observations) are the photodetector voltages due to σ^+ and σ^- components of the probe beam. The pump-duration is ∼40 ms during which we see a sudden jump of the signal due to pump beam leaking into the photodetectors. The change in transmission is due primarily to bleaching by the pump beam (the signals return to their original levels after 30 s, as shown in 4.17). The pump beam power and diameter were $\sim$ 2 mW and $\sim$ 0.5 cm respectively. The probe beam power and diameter were $\sim$ 3 μW and $\sim$ 0.4 cm respectively.

example of timing control in our experiment. First, data is taken with pump beam off for 100 ms. Then the pump beam is turned on for 40 ms after which it is turned off for the rest of the duration of that measurement.

The basic idea behind optical pumping and T_1 time has been explained in section 1.3. When the magnetic field is transverse (that is when the applied longitudinal field along the direction of pump beam is off), a rapid precession of the atoms occur and therefore the net polarization formed by the pump beam is too low to observe. When the magnetic field is applied longitudinally in the direction of pump beam a net spin polarization in the atomic ensemble of rubidium is created due to optical pumping, as explained in chapter 1. Then we should be able to observe differential absorption

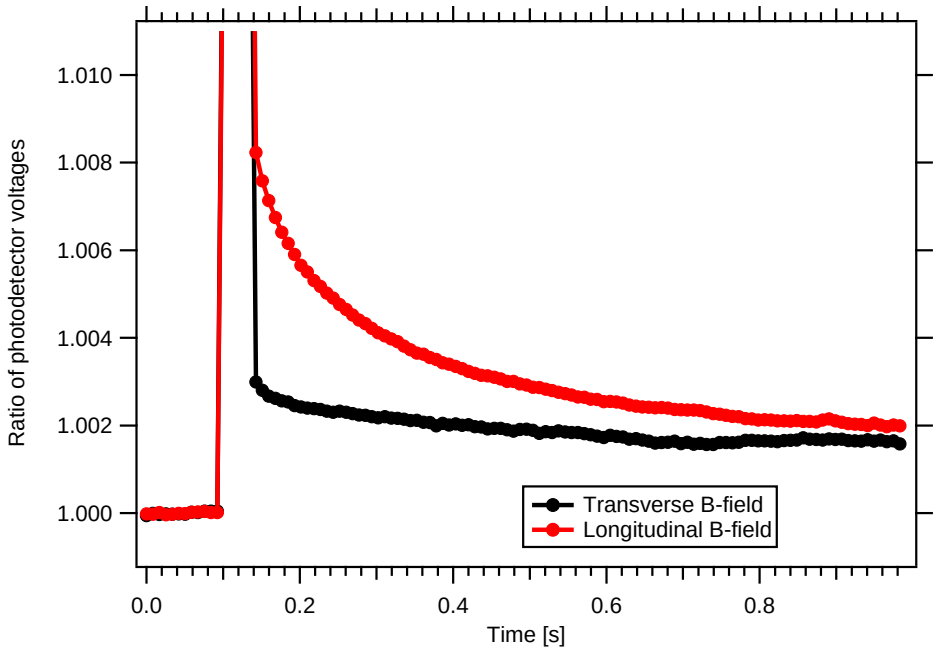


Figure 4.13: Ratio of photodetector voltages due to σ^+ and σ^- components of the probe beam. The black curve is obtained with transverse magnetic field, and the red curve is obtained with longitudinal magnetic field in the direction of pump beam. This red curve is obtained from the raw data of figure 4.12 in which the overall bleaching is much larger than the difference in the ratios of the σ^+ and σ^- signals seen in this figure.

of σ^+ and σ^- components of the probe beam during the pumping period. When the shutter is closed the pump beam is blocked and there is no more pumping of rubidium atoms. Then the polarization vector starts to relax to thermal equilibrium, and we should again be able to observe decaying differential absorption of σ^+ and σ^- and measure the time duration (T_1) for relaxation. Unfortunately, due to a leak of the pump beam to the photodetectors we are unable to accurately detect differential absorption of σ^+ and σ^- components of the probe beam during the pumping period. However, we are able to detect such differential absorption and measure relaxation time T_1 after the pump beam is off.

The raw data is presented in figure 4.12 which shows photodetector voltages due to σ^+ and σ^- components of the probe beam. To detect the change in polarization

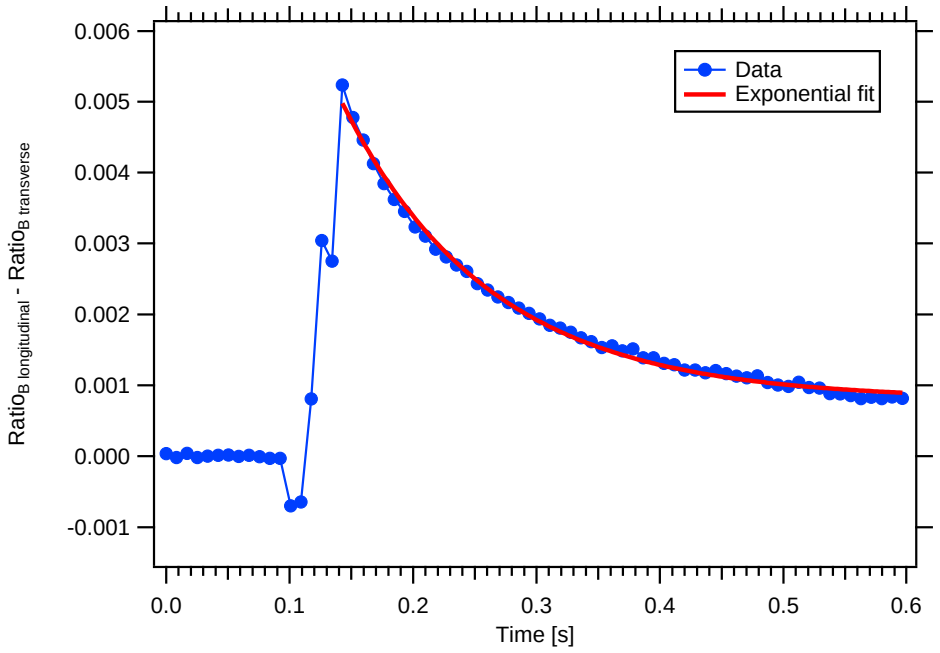


Figure 4.14: The difference of the ratio of photodetector voltages (obtained with applied longitudinal magnetic field) plotted over time. The red curve is the exponential fit (with offset) from which we can extract T_1 time. The graph is obtained from the ratios (red and black curves) of figure 4.13.

we find the ratio of these voltages. We find such ratios with both longitudinal and transverse magnetic field. Figure 4.13 presents these ratios (normalized with the ratios before pumping) plotted over time from a typical pumping and relaxation process. In the figure we can see that before the pump beam is turned on, the ratios with and without magnetic field are the same. When the pump beam is turned on we see a sudden jump of signal due to a leakage of pump beam to the photodetectors. When the pump beam is turned off the signal to noise ratio is good enough to observe the relaxation of polarization. The black curve in figure 4.13 is with transverse magnetic field. It is not known why the pumping beam also slightly changes the signal level with transverse magnetic field.

We find the difference of the ratios with longitudinal magnetic field and with transverse magnetic field, and then fit the difference with the function 4.1 to extract

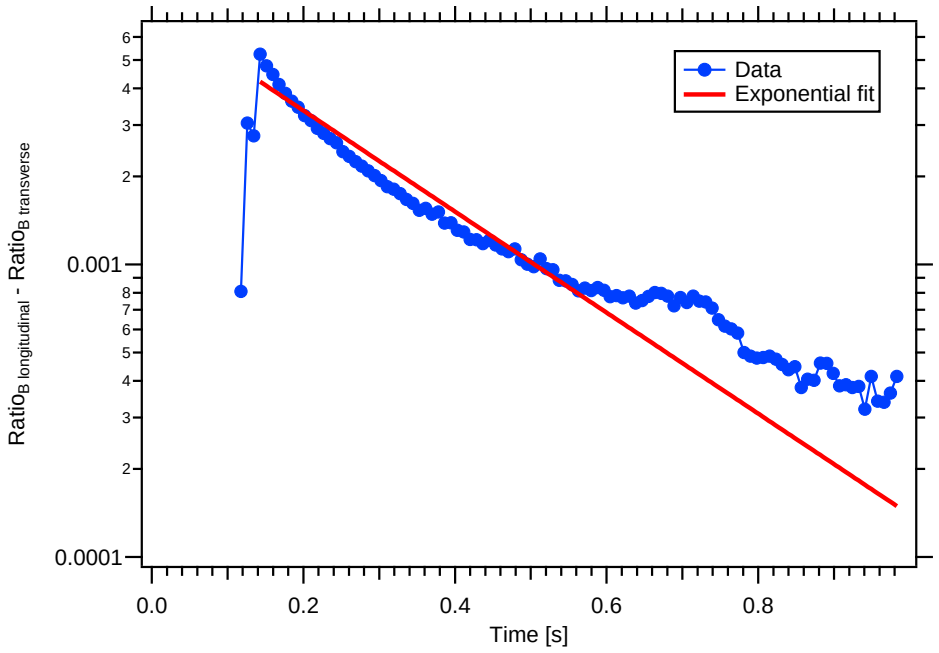


Figure 4.15: The difference of the ratio of photodetector voltages (obtained with applied longitudinal magnetic field) in log scale plotted over time. The red curve is the single exponential fit (without offset) which does not fit well with the data over duration of 0.8 s. Compare with figure 4.16.

T_1, as presented in figure 4.14. For the particular data set shown in the figure 4.14 the fitting was done from initial decay to 0.6 s, and the value of T_1 time was found to be 120 ms. In our experiment, the values of T_1 range from 50 to 200 ms depending upon the conditions of pumping. We have found that the data fit better with double exponential ($y = A_1 e^{-x/\tau_1} + A_2 e^{-x/\tau_2}$) than with single exponential function ($y = Ae^{-x/\tau}$), as shown in figures 4.15 and 4.16. Therefore the values of T_1 obtained by using single exponential fit are the most pessimistic ones (giving lower limit of T_1).

In the figure 4.12 we can see that once the pump beam is off and polarization relaxes, the signals do not come back to the original level (to the level before pump was turned on). This is due to bleaching as discussed in 4.1. However, when we increased the duration of data taken after the pump beam is off, we see that the signals (for both magnetic field on and off) come back to the original level. Figure 4.17 shows that the signals come back to original level, due to unbleaching, after

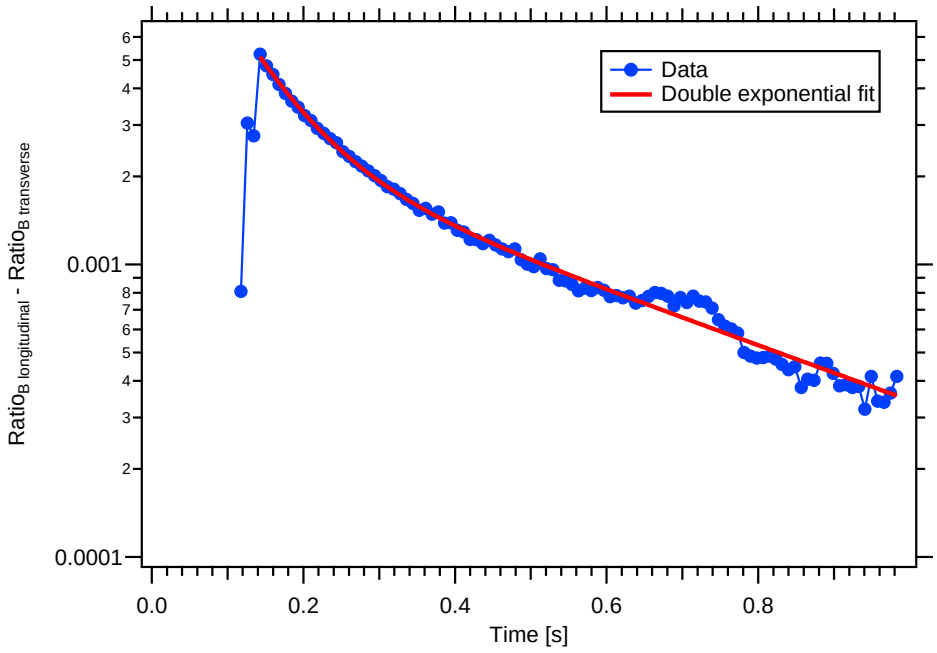


Figure 4.16: The difference of the ratio of photodetector voltages (obtained with applied longitudinal magnetic field) in log scale plotted over time. The red curve is the double exponential fit (without offset) which fits well with the data over duration of 0.8 s. Compare with figure 4.15.

about 30 seconds.

4.2.2 Duration of pumping and its effect

As discussed in the section 4.1 the higher the pump power the higher the bleaching effect for a given pumping duration. The bleaching problem can also be controlled by reducing the duration of pumping for a given pump power.

Figure 4.18 shows the the raw data obtained from two photodetectors looking at σ^+ and σ^- components of probe beams. As we can see the longer the duration of pumping (for a given pump power) the higher the bleaching, and therefore it takes longer to come back to the original level. Figure 4.19 is obtained from the data of figure 4.18 using lamp as an unbleaching light source. We find the difference of the voltage before pumping and the voltage just after pumping is off, and then plot this

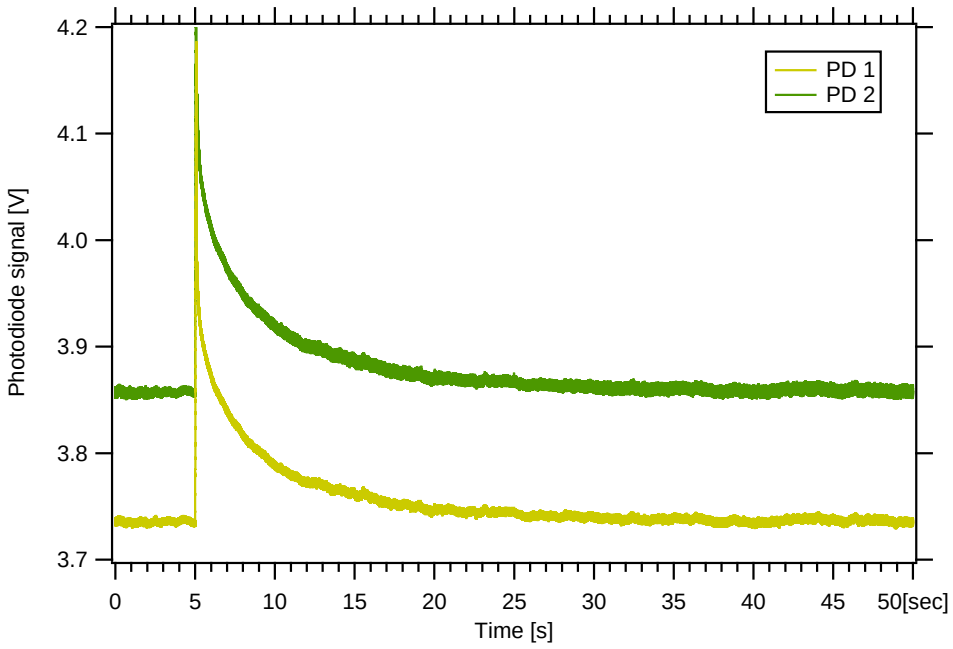


Figure 4.17: Raw data taken over 50 s. The green and golden curves are the photodetector voltages due to σ^+ and σ^- components of the probe beam, which come back to their original levels after about 30 s.

difference against the pumping duration. We can see that the plot is of exponential increasing form. Fitting with the function 4.1 we find the value of τ to be ∼110 ms. For the given conditions (pump power of $\sim$ 2 mW and diameter $\sim$ 0.5 cm), the pumping duration should be below this value in order to mitigate bleaching to measure T_1 time.

We did not observe any dependence of T_1 time with the pumping duration for a given intensity of the unbleaching light source. This has been presented in figure 4.20, where we can see that T_1 time ranges from 140 ms to 190 ms, and is more or less within this range for all pumping durations.

However, the change in the amplitude A of the fit function 4.1 with pumping duration can be observed, as presented in figure 4.21. Similarly, the difference of ratios (of σ^+ and σ^-) with magnetic field on and off also changes with pumping duration, which is shown in the same figure 4.21. Based on these observations we can see that

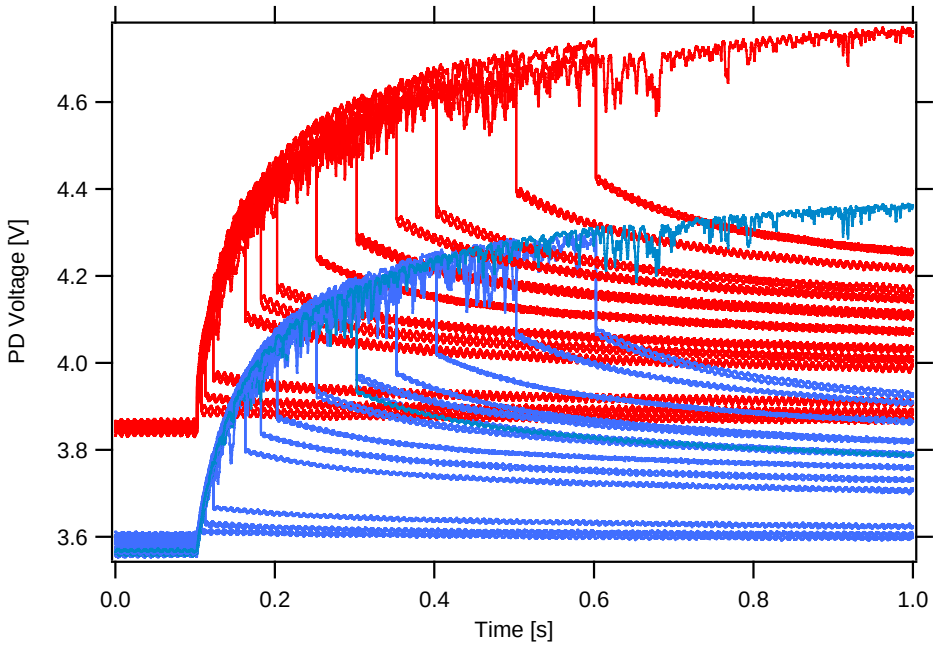


Figure 4.18: Raw data taken with different pump-durations for a fixed pump power. The red and blue curves are the photodetector voltages due to σ^+ and σ^- components of the probe beam.

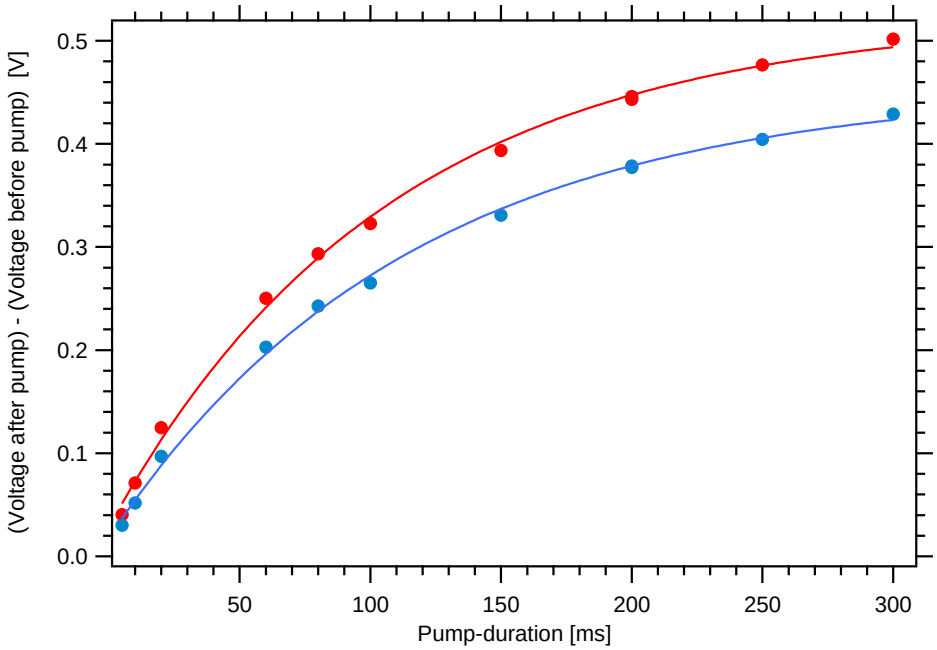


Figure 4.19: Difference of photodetector voltages before and just after pumping plotted against pump-durations. Taken from raw data of figure 4.18.

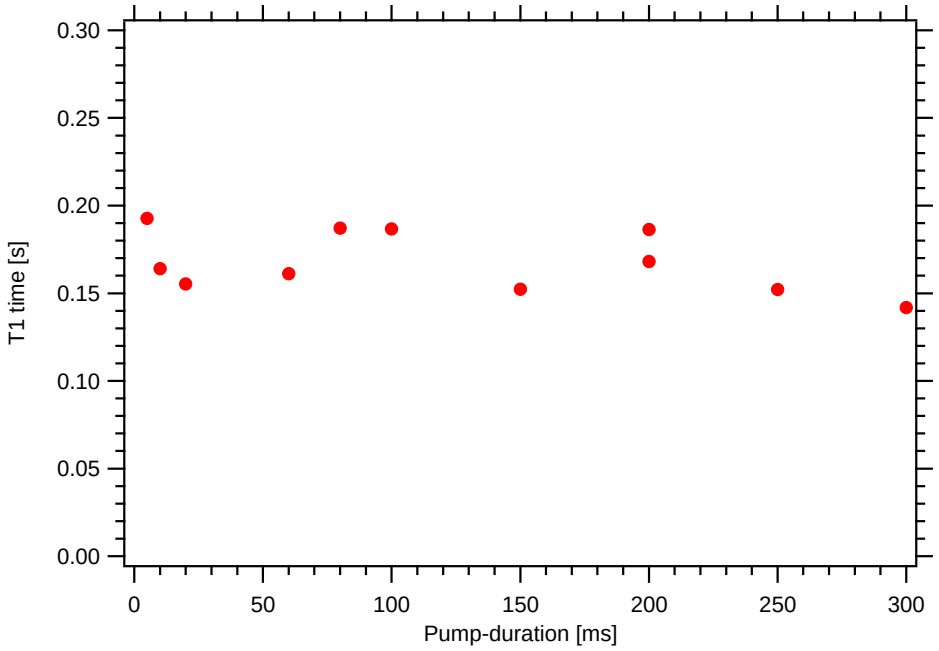


Figure 4.20: T_1 time vs. pump-duration. The pump beam power and diameter were ~ 2 mW and ~ 0.5 cm respectively. The probe beam power and diameter were ~ 3 μW and ~ 0.4 cm respectively.

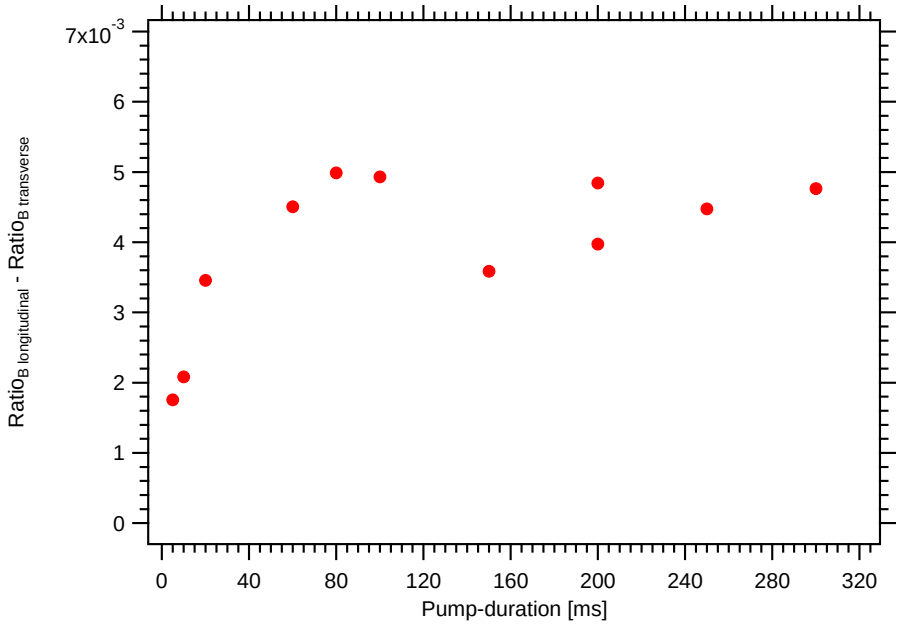


Figure 4.21: The difference of ratio of σ^+ and σ^- components of probe beam with longitudinal and transverse magnetic field, plotted against the pump-durations.

the pumping duration causing the highest amplitude and the biggest difference of ratios is between 60-100 ms for the given conditions (lamp as a light source, probe beam intensity of $\sim$0.01 mW/cm^2 , pump beam intensity of $\sim$2 mW/cm^2 at the Ar-Rb crystal).

4.2.3 Effect of temperature on the Ar-Rb crystal

We expect that the relaxation time T_1 depends upon the temperature of Ar-Rb crystal. We took data (similar to those discussed in subsection 4.2.1 at different temperatures of the Ar-Rb crystal from 3.2 K to 5.2 K. Figure 4.22 show that T_1 time decreases as the temperature of Ar-Rb crystal is increased, although the data is noisy. Additionally, figure 4.23 shows that the amplitude A (extracted from data as discussed in subsection 4.2.1) of the fit (equation 4.1) also decreases with the temperature of the crystal. This shows that if we could reduce the temperature of Ar-Rb crystal further below 3.2 K we can improve T_1 time. Unfortunately, with the current setup of our cryostat the temperature is limited to 3.2 K.

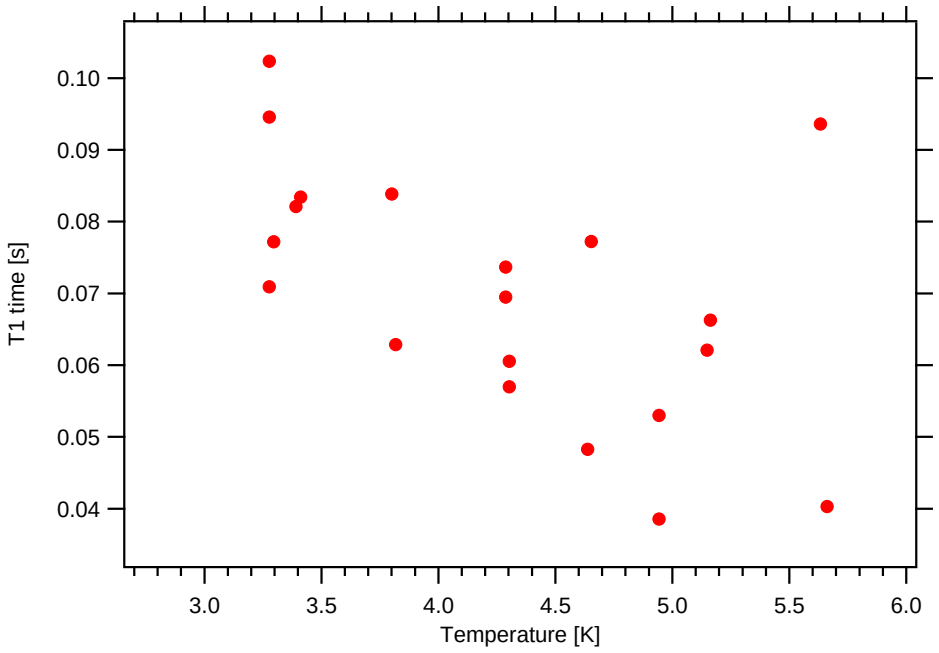


Figure 4.22: T_1 time vs. temperatue of Ar-Rb crystal

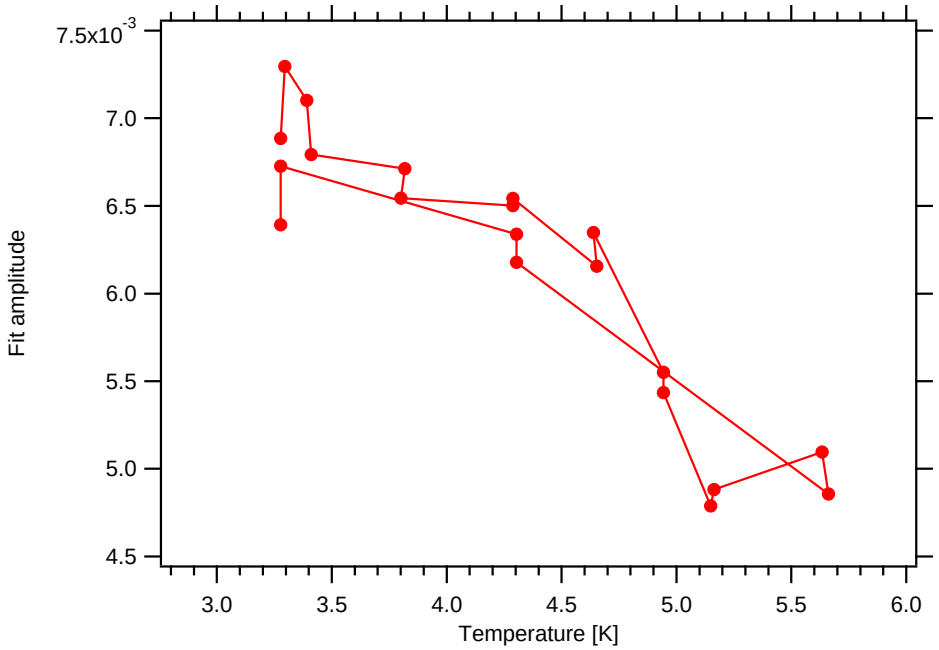


Figure 4.23: The amplitude of the exponential fit (equation 4.1) vs. temperatue of Ar-Rb crystal. The amplitude values are obtained by fitting the ratios of σ^+ and σ^- components of probe beam with longitudinal magnetic field.

Chapter 5

Summary and Future Prospects

We have trapped rubidium atoms inside a cryogenic argon crystal with a rubidium density exceeding 10^{16} cm^{-3}. The Ar-Rb crystal was grown on a sapphire substrate at 15 K and was cooled down to 3.2 K by a pulse tube cooler. Crystal growth was monitored by thin-film interferometry and the rubidium density in the argon beam was monitored by absorption spectroscopy. The crystal thickness obtained by growing it over 3.5 hours was 200 microns. After cooling down to 3.2 K three sharp peaks were observed at 799 nm, 785 nm, and 747 nm which match closely to the red triplet lines observed in past experiments [2].

After growth of the crystal we prepared Rb polarization (m-state) by optical pumping with a σ^+ laser pulse of duration $\sim$40 ms tuned at wavelength 799 nm. We measured the differential absorption of σ^+ and σ^- components of a probe laser when the pump beam is off, from which we extracted the polarization relaxation time (T_1) of about 0.1 s. We also observed an undesirable optical pumping effect ("bleaching") which caused significant reduction of the OD of the three red peaks; however, we found that the bleaching effect can be mitigated by applying an additional light of wavelengths in the range of 660-730 nm.

There are many things to explore in our future work. Our experiment was limited

to a measurement of the T_1 time. In future, we can improve the experiment to measure the T_2 time. Additionally, we can explore the dependence of the T_1 and T_2 times with temperature and rubidium density.

We can also explore matrices of neon and molecular hydrogen, and different other dopant atoms such as cesium (large fine structure of Cs could be favorable for optical spin detection). Neon has advantage of having weaker bonds than argon. H_2 has bonds weaker than noble gases, and therefore provides narrower lines and also helps dopants to retain their gas-phase properties [62]. It can be prepared in the parahydrogen state having $I = 0$. It is known that with parahydrogen crystals of high optical quality can be grown up to 1 mm in thickkness [63]. There have been many experiments of isolation of H, Li, B, N, Na, Mg, and Al atoms in solid hydrogen [64, 65]. Lithium atoms have been isolated in mixture of para- and ortho-hydrogen and studied spectroscopically [66]. With parahydrogen narrower lines are expected [67]. These systems will be explored in future work.

Appendix A

Mechanical Drawings

This section presents all mechanical drawings for parts designed for the experiments. Drawings were created using Alibre Design™ software.

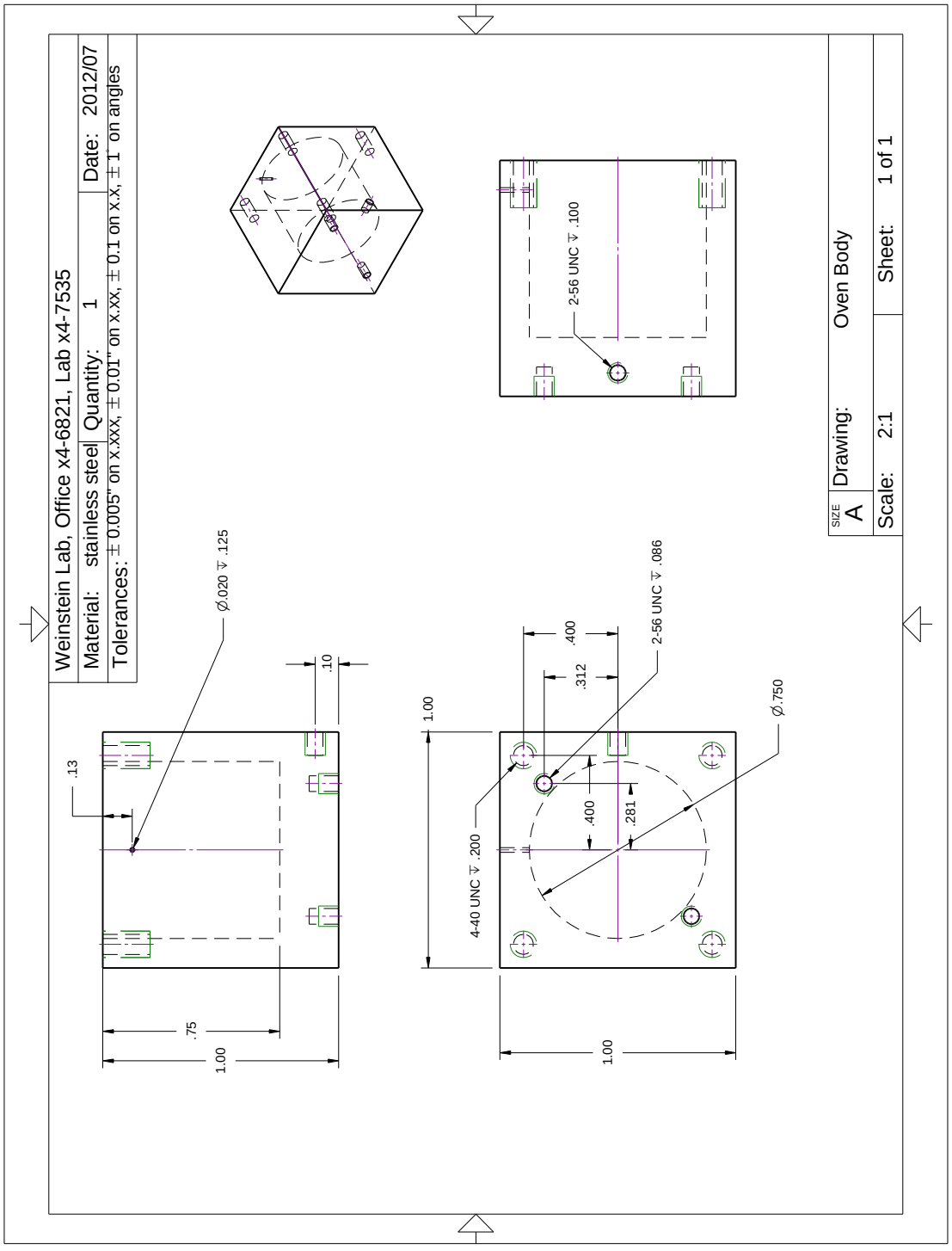


Figure A.1: Oven Body

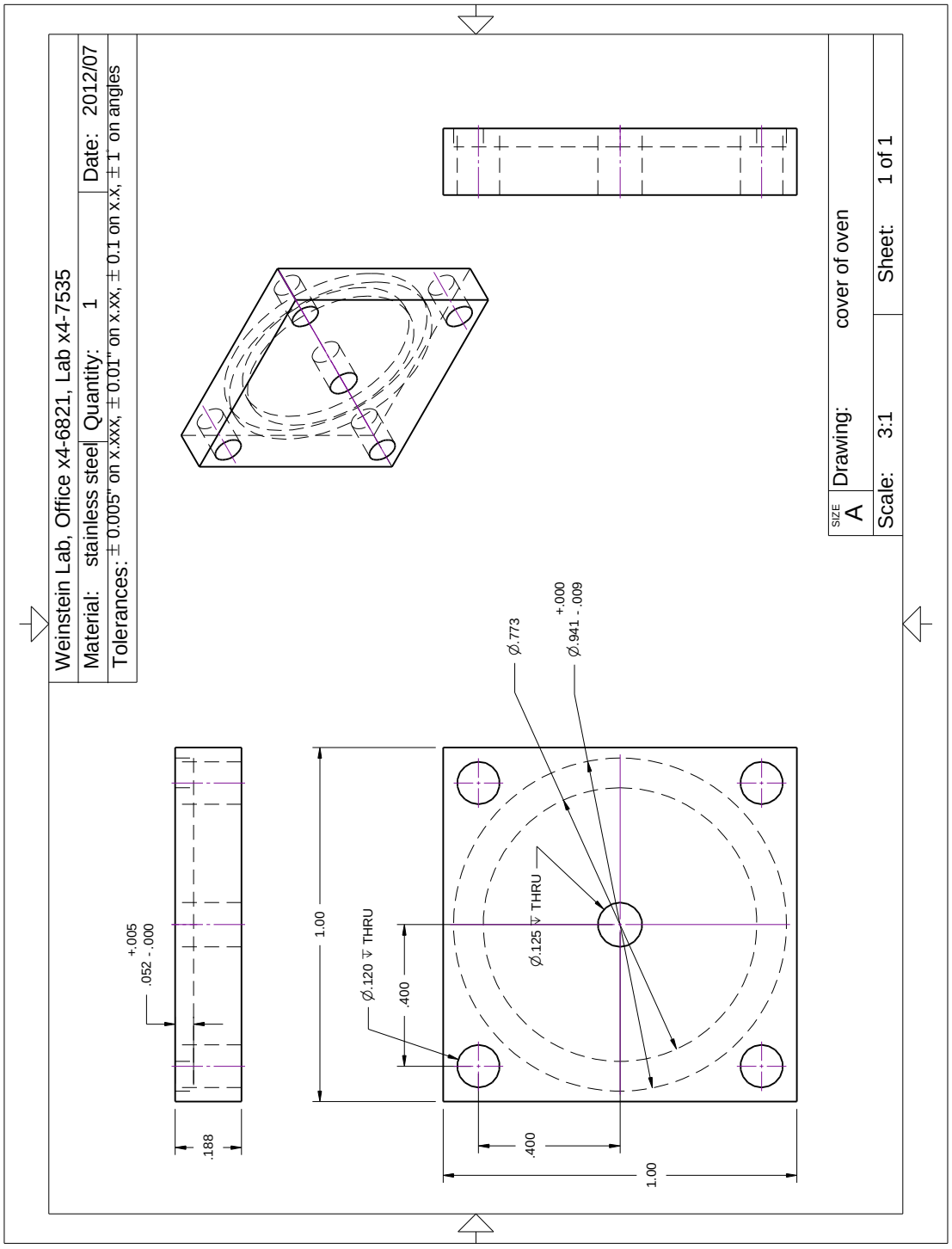


Figure A.2: Oven Cover

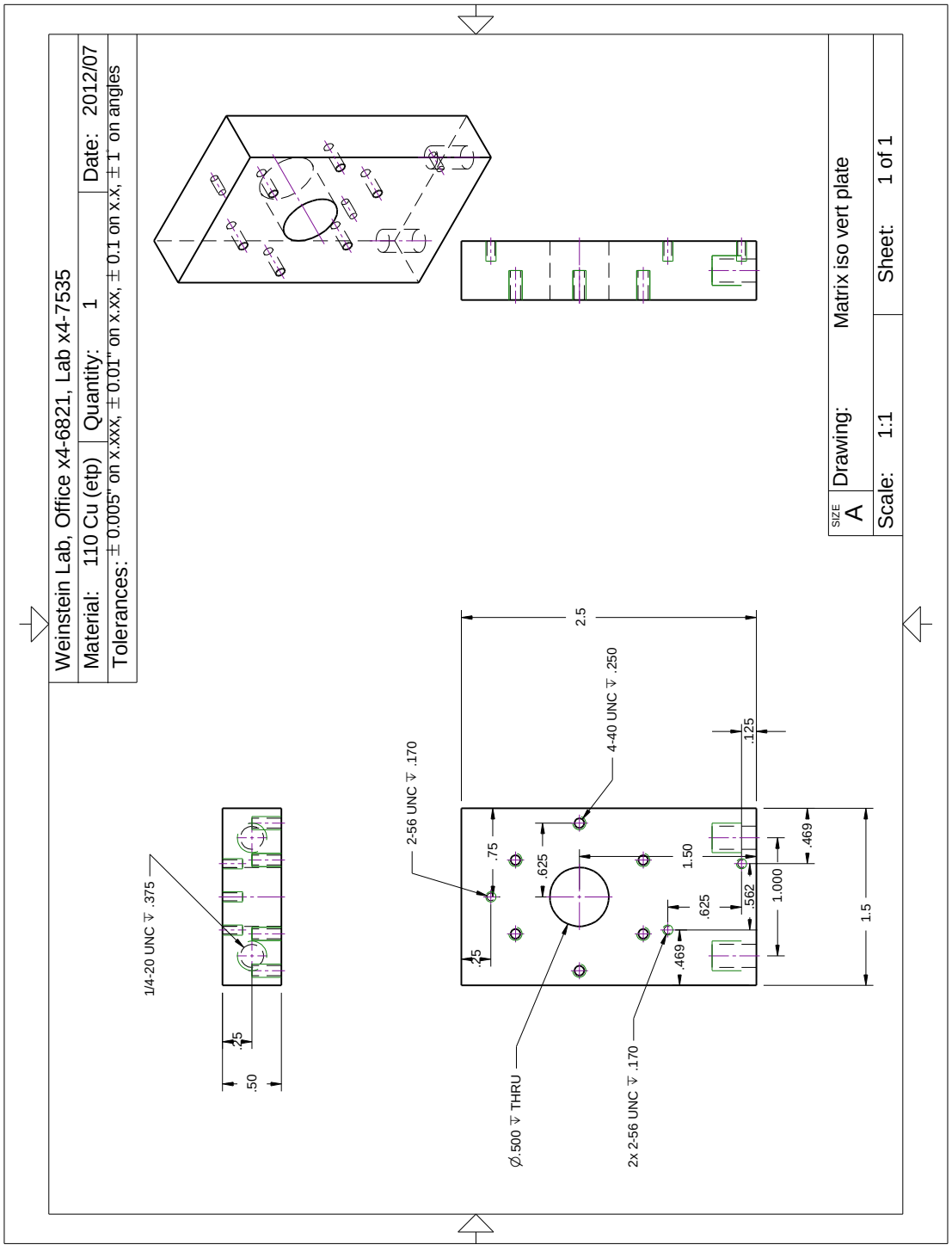


Figure A.3: Vertical Plate

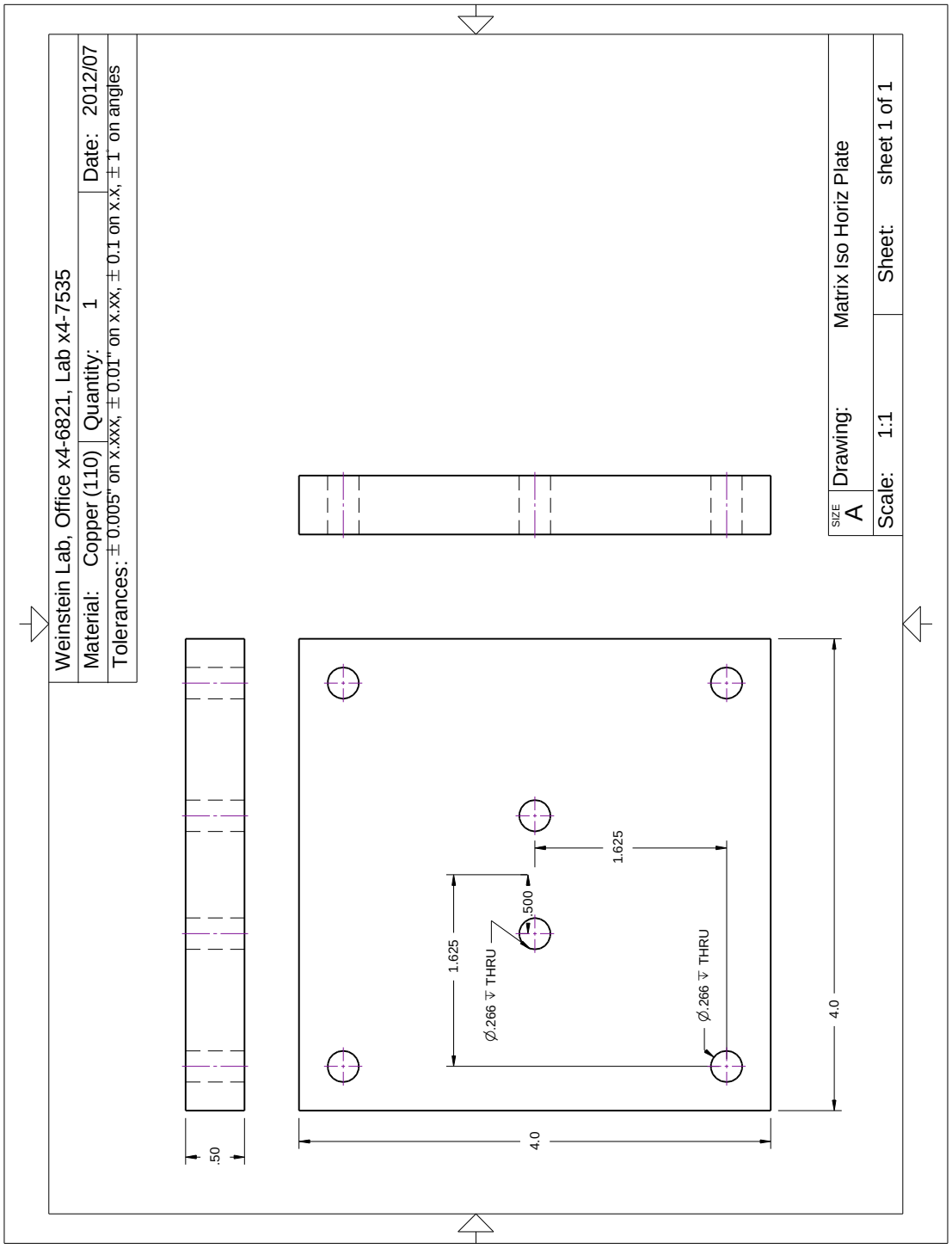


Figure A.4: Horizontal Plate

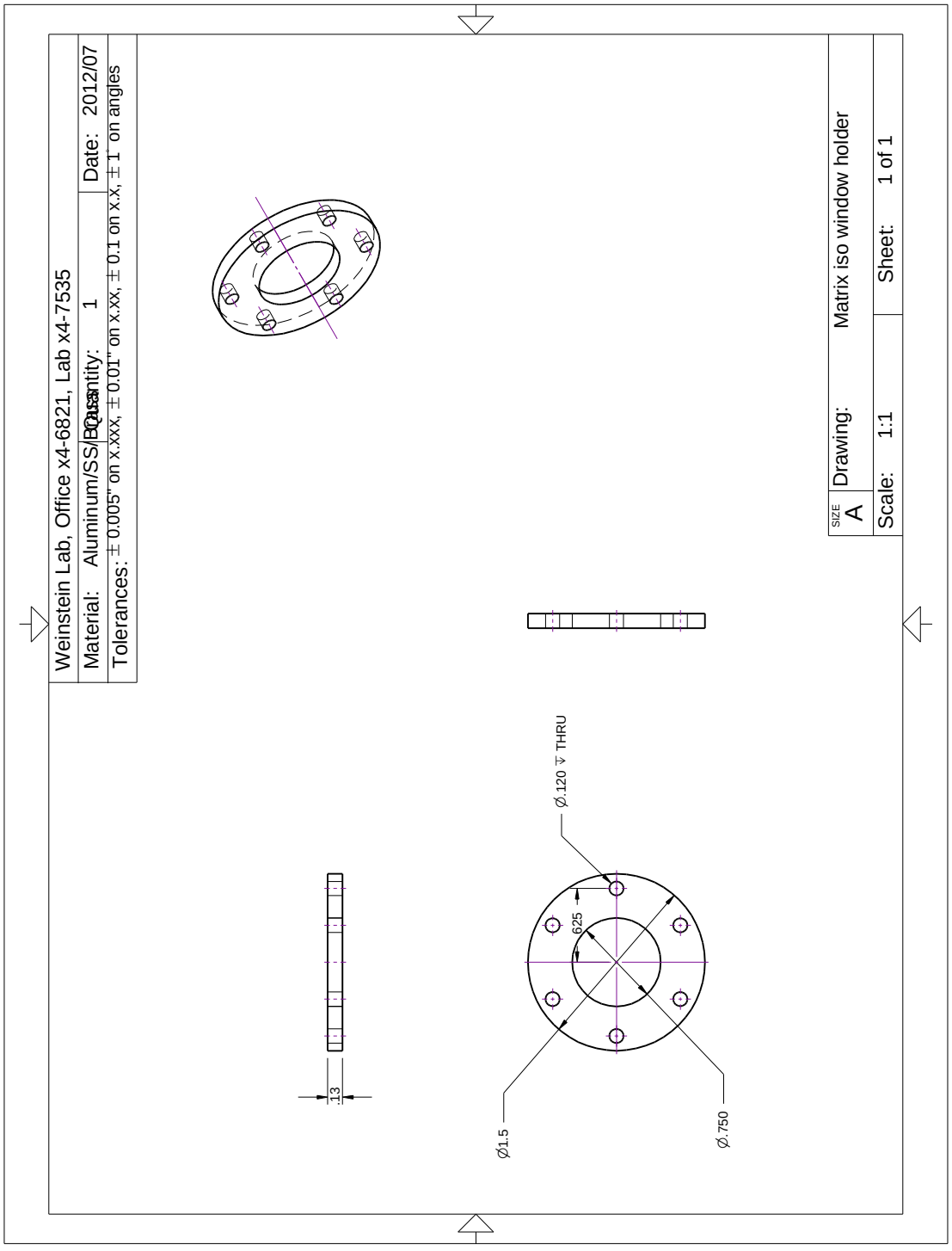


Figure A.5: Sapphire Window Mount

Appendix B

Rubidium Vapor Pressure Curve

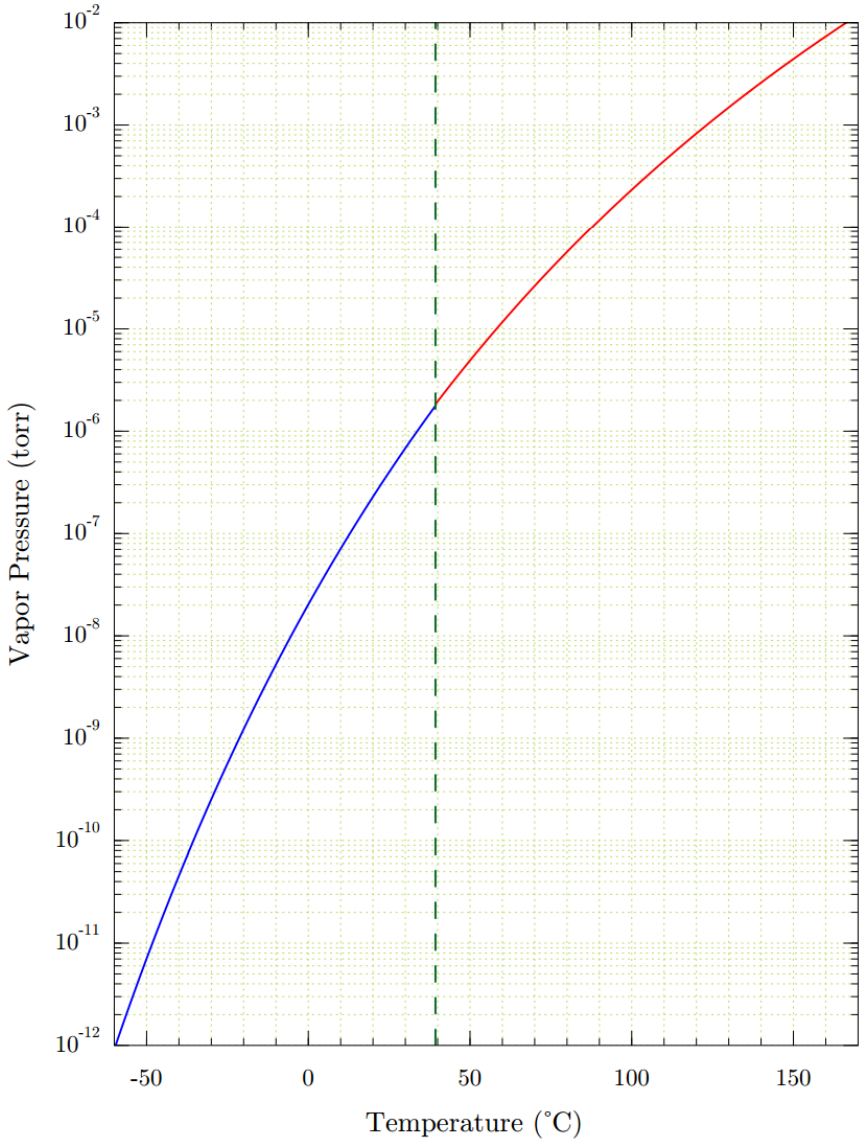


Figure B.1: Vapor pressure of rubidium vs. temperature. The vertical line seen near 40 °C indicates melting point of rubidium. Figure reproduced from the reference [3].

Appendix C

Rubidium Hyperfine Structure

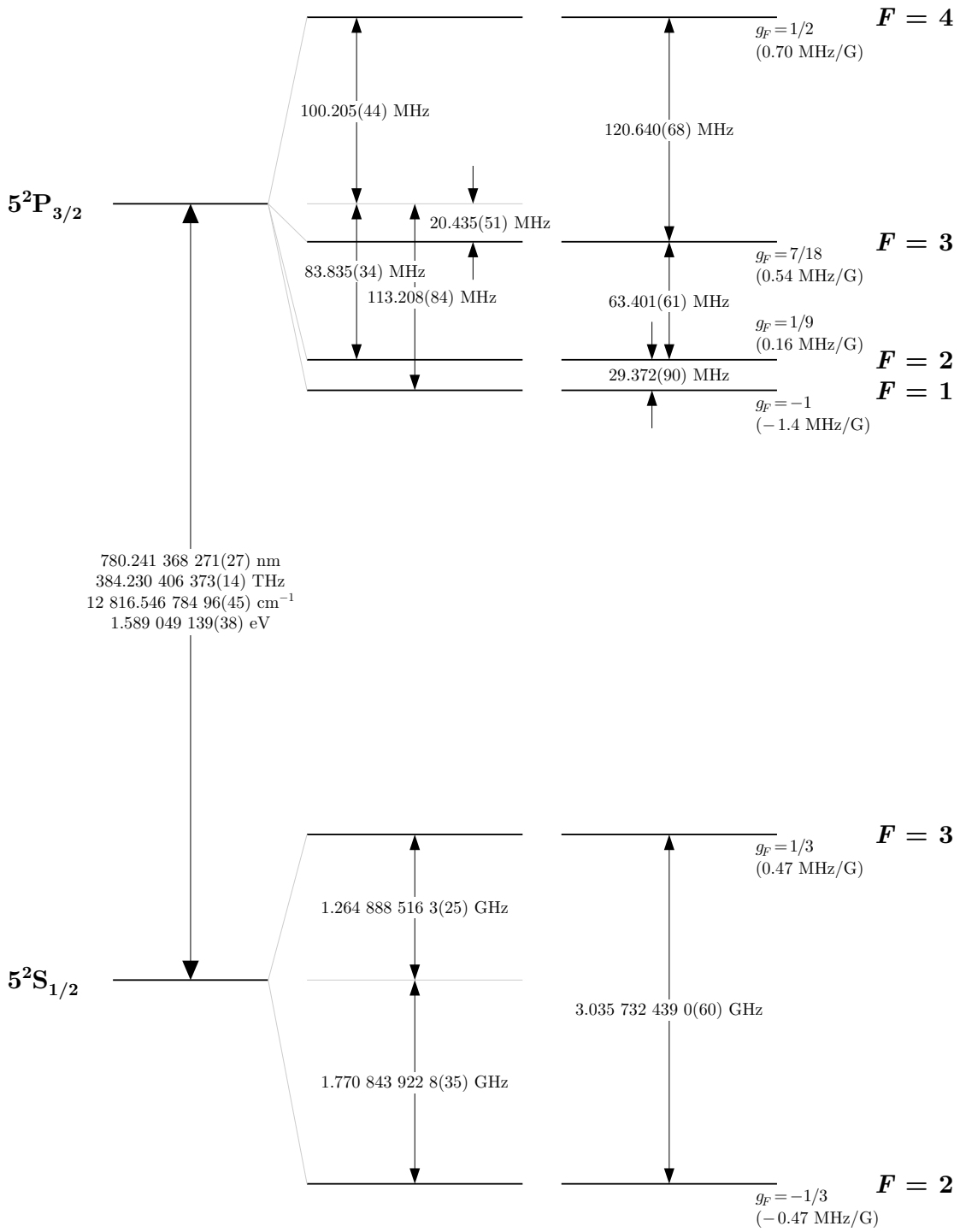


Figure C.1: ^{85}Rb D_2 transition hyperfine structure. Figure reproduced from the reference [3] where the ground-state values were obtained from [4] and the excited-state values were obtained from [5, 4].

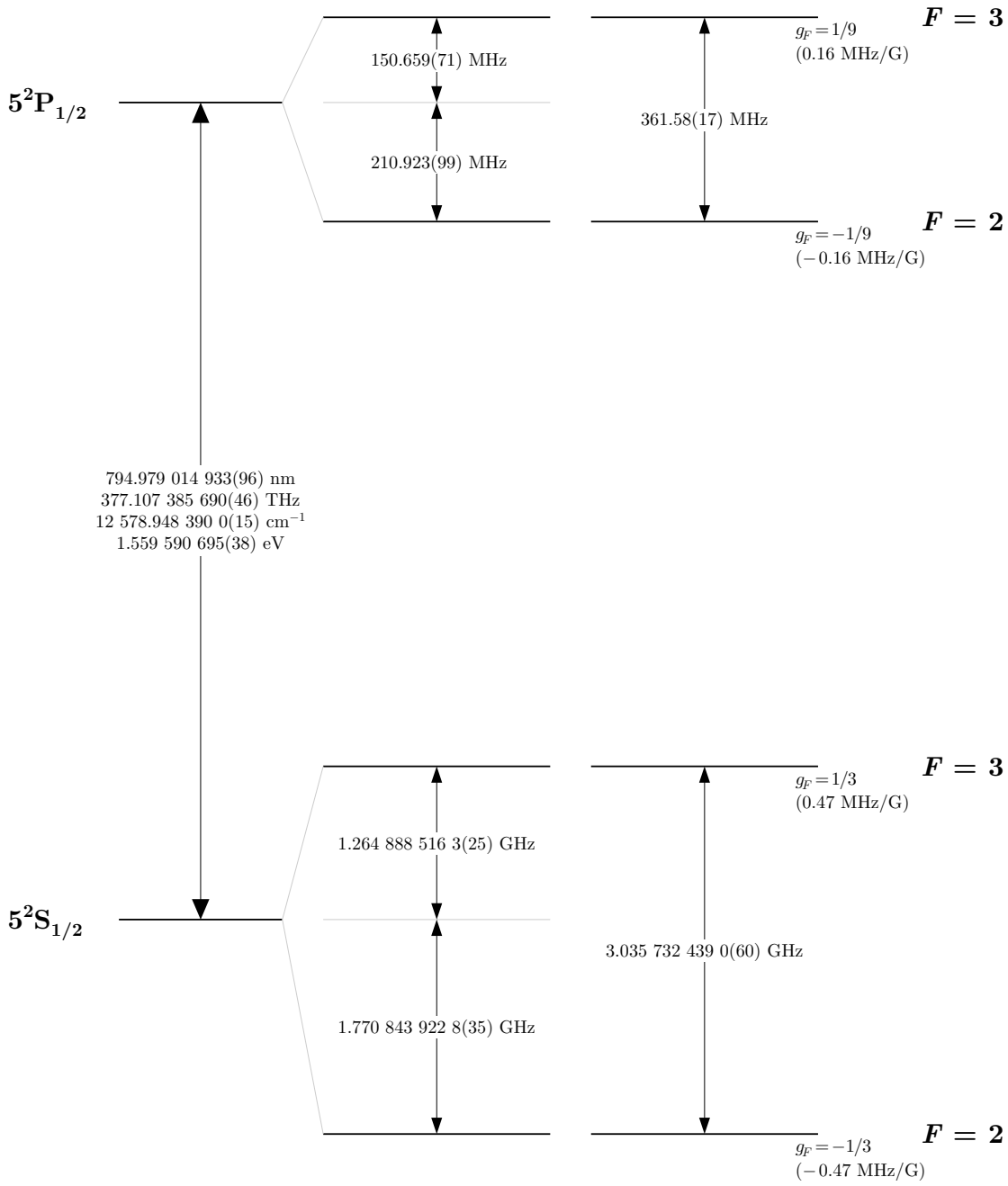


Figure C.2: ^{85}Rb D_1 transition hyperfine structure. Figure reproduced from the reference [3] where the ground-state values were obtained from [4] and the excited-state values were obtained from [6, 4].

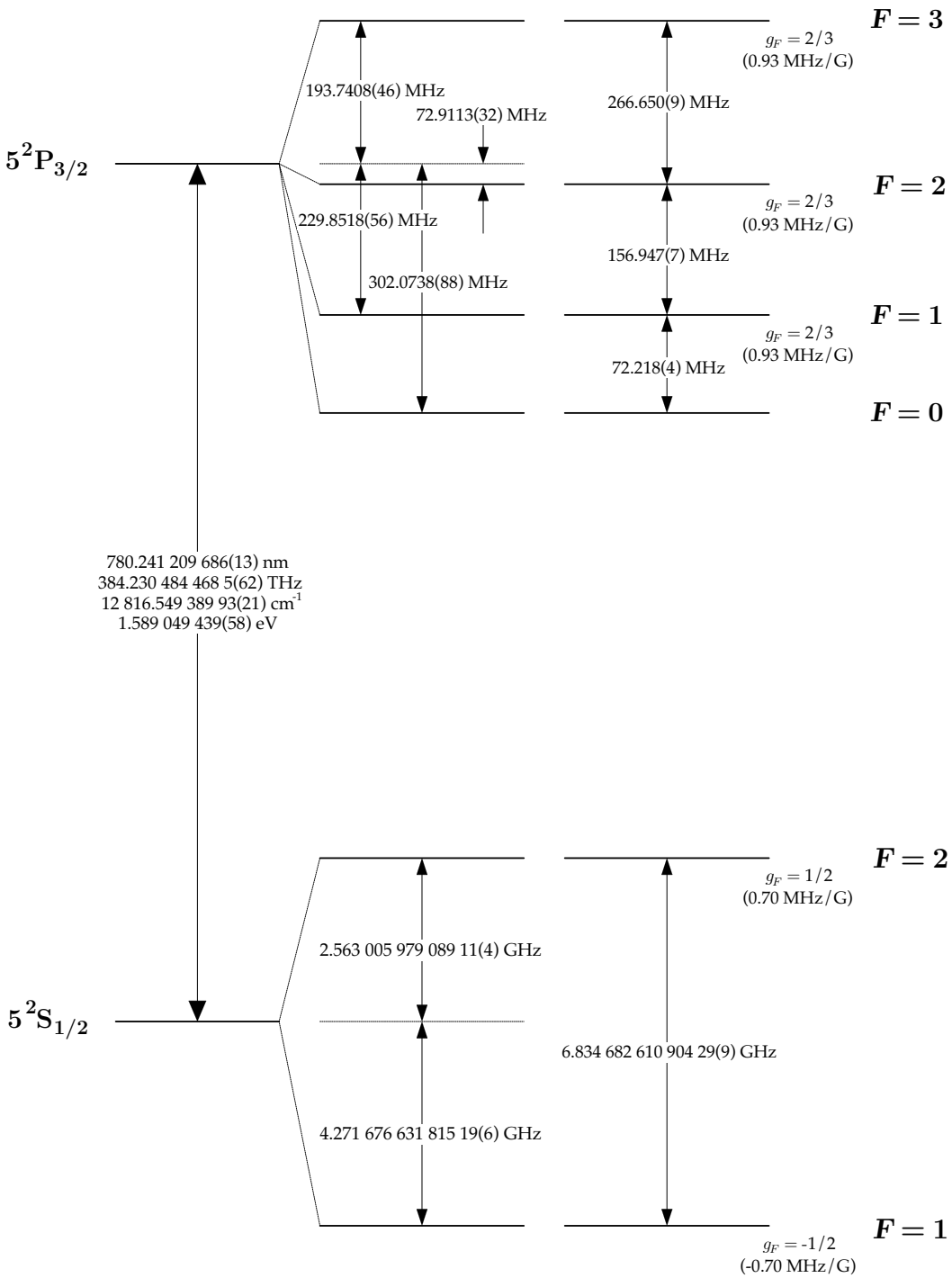


Figure C.3: ^{87}Rb D_2 transition hyperfine structure. Figure reproduced from the reference [3] where the ground-state values were obtained from [7] and the excited-state values were obtained from [8].

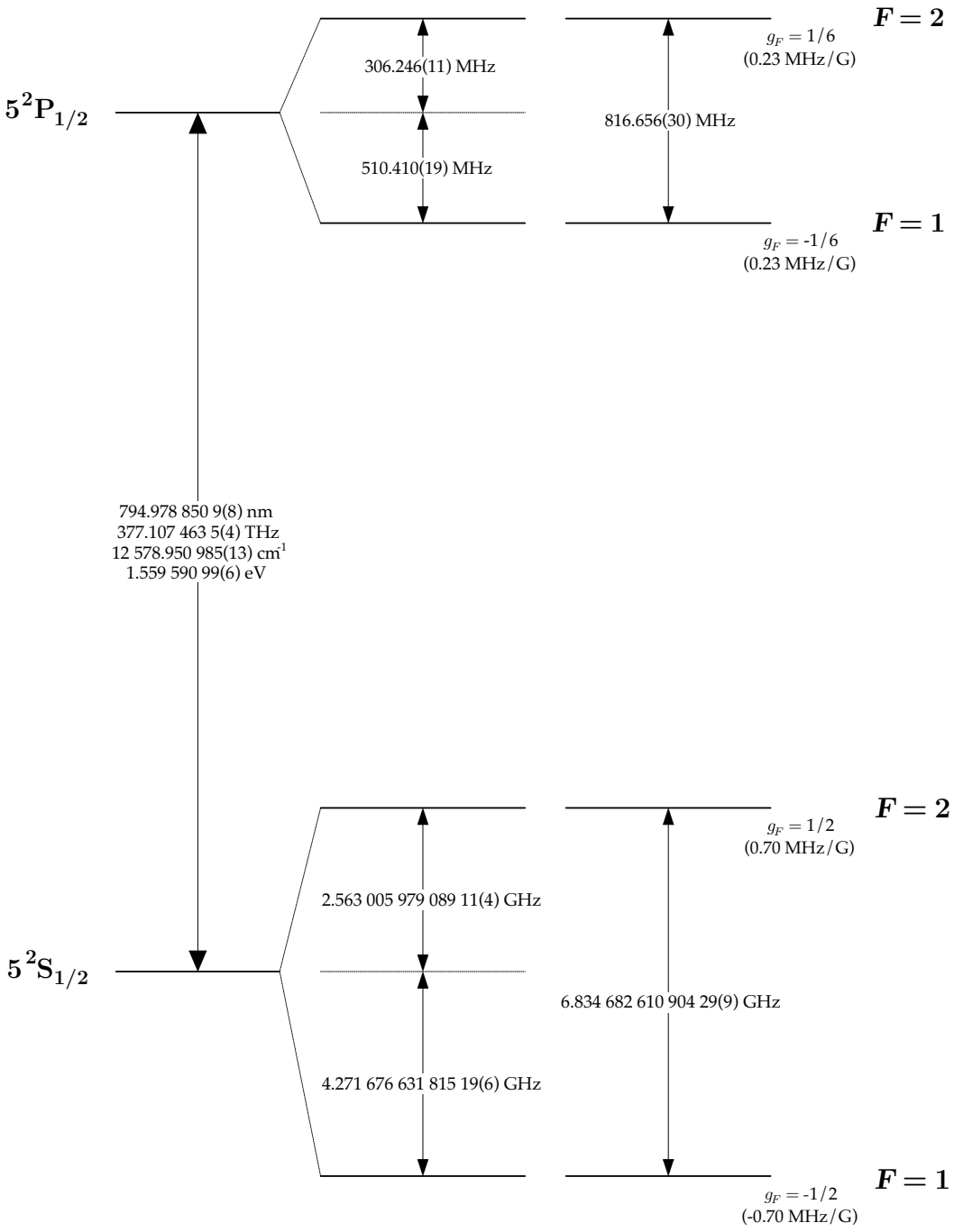


Figure C.4: ^{87}Rb D_1 transition hyperfine structure. Figure reproduced from the reference [3] where the ground-state values were obtained from [7] and the excited-state values were obtained from [6, 5, 4].

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